

AKADEMIKA

BEL-PET

POWER ELECTRONICS TRAINER

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. BASIC ELECTRONICS is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that We have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademia Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademia Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademia Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademia Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademia Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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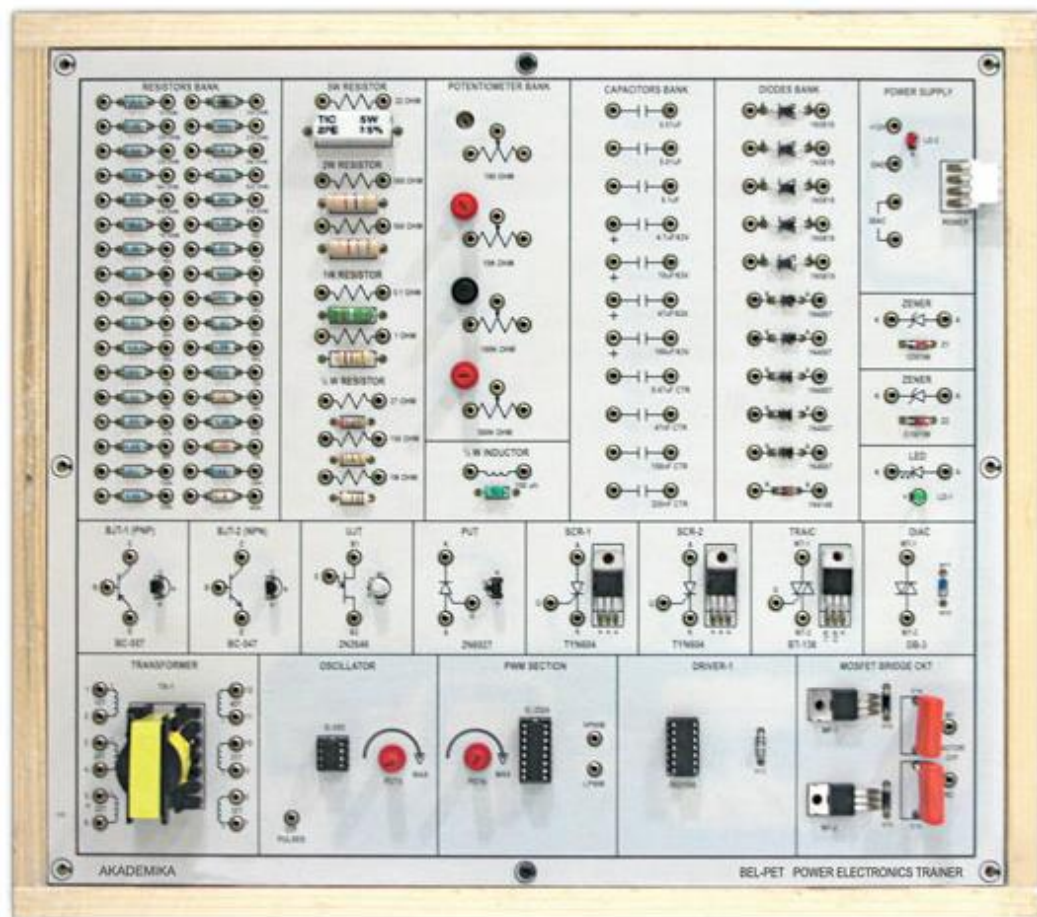
TECHNICAL SPECIFICATIONS

- Characteristics study of different types of power
- AC supply, ranging from 0~30VAC/50Hz, DC supply of +12V
- electronics Devices i.e. thyristors (SCR, TRIAC, DIAC), UJT,
- MOSFET, etc.
- Application study of different types of power electronics
- Devices i.e. thyristors (SCR, TRIAC, DIAC) and MOSFET
- Different gate-pulsing circuits (turn-on methods)
- Different commutation circuits (turn-off methods)
- Different IC's for gate triggering circuits
- Provision of freewheeling diodes
- Resistor bank
- Capacitor bank
- Inductor bank
- Potentiometer bank
- Interconnection points
- Interconnection cords
- **Accessories**

BEL-PET			Quantity
1	Short Links PCS2 Patch cord	Male To Male	15
2	Long Links PCS2 Patch cord	Male To Male	5
3	EXPT Manual		1
4	PET Power Supply		1
5	Power cord		1
6	Single Strand Wire 2mtr		1

Note: Reading Tables and Graphs provided in this manual are taken on sample trainer board hence actual readings and graphs may differ from the readings and graphs shown in this manual. Readings and graphs in this manual are to be treated as sample only for reference purpose to get idea of characteristics of the respective component.

FUNCTIONAL BLOCK



INTRODUCTION

Akademika Lab Solutions' BEL-PET Power electronics Trainer kit" is a compact single board unit in wooden casing it provides hands on experimentation in the area DC to AC converter, DC to DC converter, Diac, Triac, etc. It broadly deals with controlling the flow of electrical power using electronic switching devices.

This kit it is possible to embark on any formative path, in fact, starting from simple experiments, it makes you grow in experience and complexity up to being able to deal with competence and master the fundamental elements which makes the Basic electronics based .

EXPERIMENT NO. 1

EXPERIMENT NO: 1

NAME:-

Simple DC to AC Inverter

OBJECTIVE:-

Study of simple DC to AC converter

THEORY:-

In one simple inverter circuit, DC power is connected to a transformer through the centre tap of the primary winding. A switch is rapidly switched back and forth to allow current to flow back to the DC source following two alternate paths through one end of the primary winding and then the other. The alternation of the direction of current in the primary winding of the transformer produces alternating current (AC) in the secondary circuit.

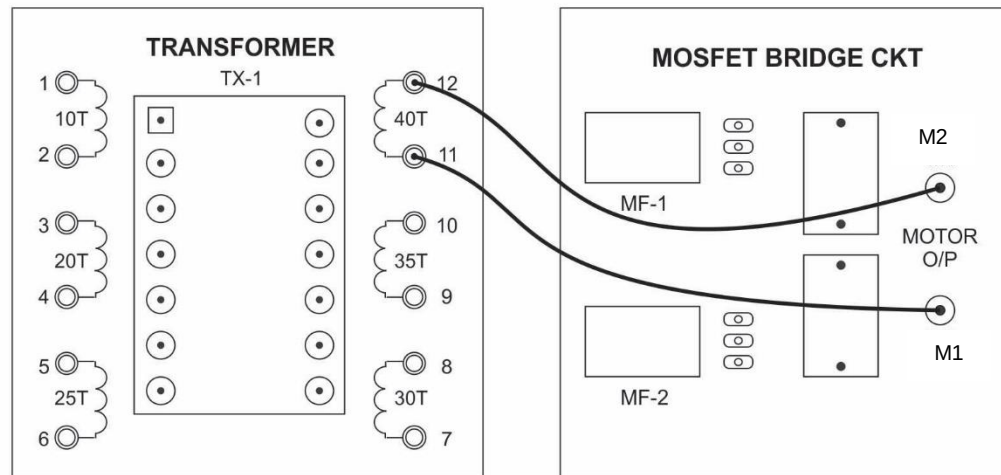
The electromechanical version of the switching device includes two stationary contacts and a spring supported moving contact. The spring holds the movable contact against one of the stationary contacts and an electromagnet pulls the movable contact to the opposite stationary contact. The current in the electromagnet is interrupted by the action of the switch so that the switch continually switches rapidly back and forth. This type of electromechanical inverter switch, called a vibrato or buzzer, was once used in vacuum tube automobile radios. A similar mechanism has been used in door bells, buzzers and tattoo guns. As they became available with adequate power ratings, transistors and various other types of semiconductor switches have been incorporated into inverter circuit designs.

PROCEDURE:-

- Make sure that power supply must be OFF.
- Refer to the circuit diagram and carry out the following connections.
- Connect power supply in proper polarity to kit BEL-PET.
- Connect M1 and M2 terminals in the MOSFET bridge circuit to 4 OT winding of transformer i.e. pin no.11 and 12 respectively, you should not connect HPWM & LPWM anywhere. It is connected internally in circuit.
- After connecting all connections power supply should be ON.
- Check the PWM pulses at HPWM and LPWM and vary the POT 6 to change the duty cycle of PWM.
- For perfect o/p keep POT 6 at Minimum position that give 50% duty cycle at o/p.
- Then observe the o/p at M1 & M2, you will get AC square wave.

- Also observe the waveform on the other winding of Transformer .you can observe same waveform in step down mode

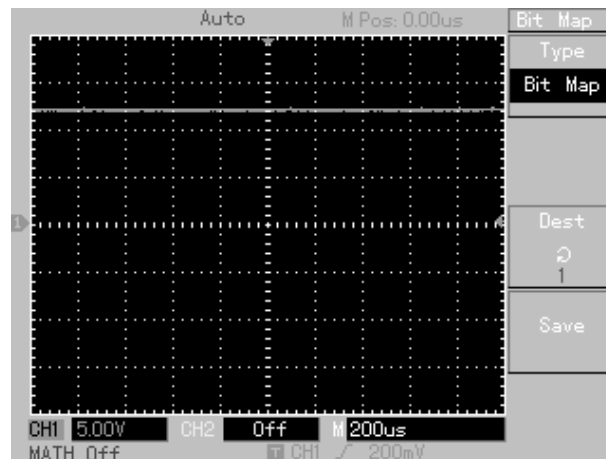
CIRCUIT DIAGRAM:-



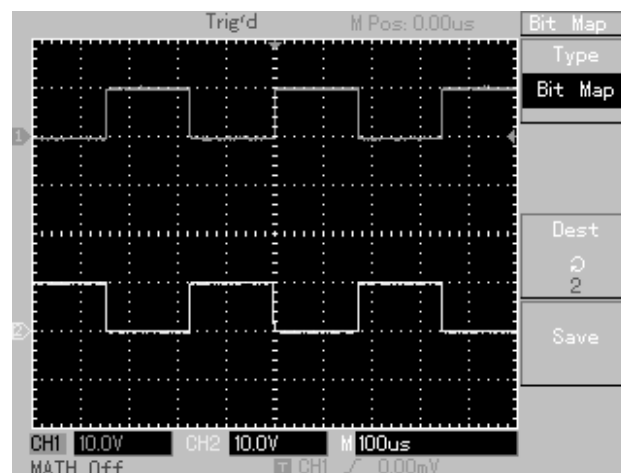
As per the above connection for 40T , you can also do same connections for different values.

OBSERVATION:-**CAPTURED WAVEFORM**

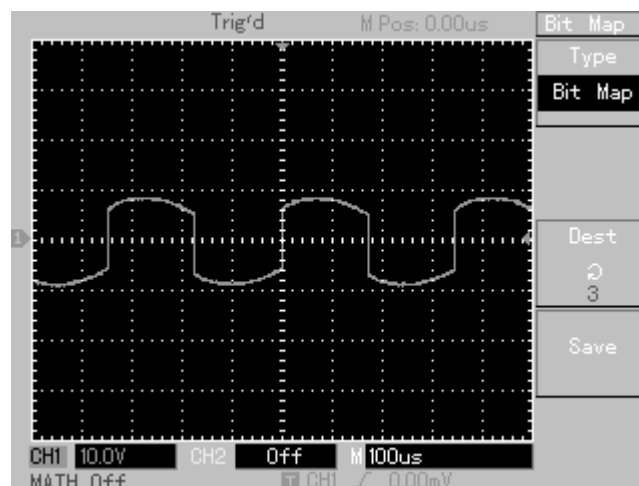
- DC Input(12VDC)



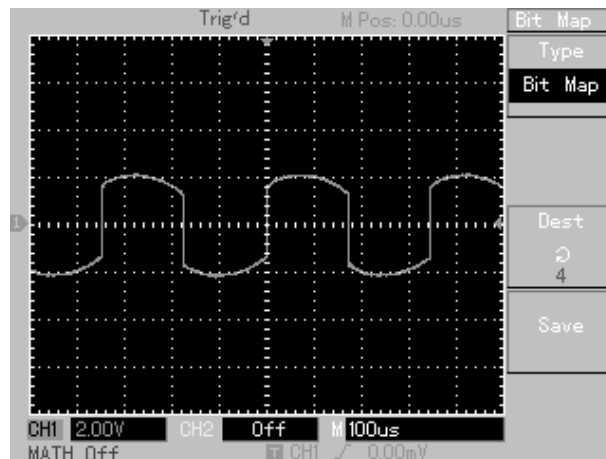
- HPWM,LPWM



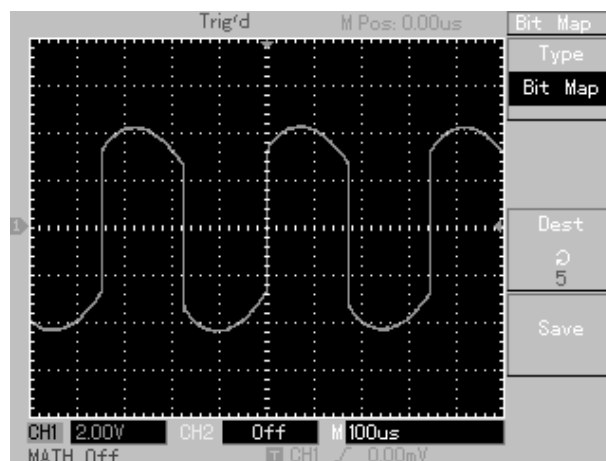
- Motor O/P M1 &M2(AC across motor O/P)



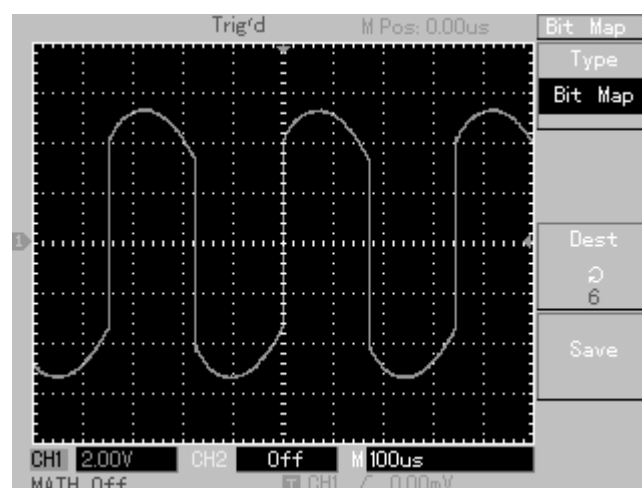
- Transformer TX1 output across 10T step down AC



- Transformer TX1 output across 20T step down AC



- Transformer TX1 output across 25T step down AC



EXPERIMENT NO. 2

EXPERIMENT NO: 2

NAME:-

DC To DC Converter (Dual Converter)

OBJECTIVE:-

Study of DC TO DC converter (dual converter)

THEORY:-

Dc-dc power converters are employed in a variety of applications, including power supplies for personal computers, office equipment, spacecraft power systems, laptop computers, and telecommunications equipment, as well as dc motor drives. The input to a dc-dc converter is an unregulated dc voltage V_g . The converter produces a regulated output voltage V , having a magnitude (and possibly polarity) that differs from V_g . For example, in a computer off-line power supply, the 120 V or 240 V ac utility voltage is rectified, producing a dc voltage of approximately 170 V or 340 V, respectively. A dc-dc converter then reduces the voltage to the regulated 5V or 3.3 V required by the processor ICs. High efficiency is invariably required, since cooling of inefficient power converters is difficult.

The ideal dc converter exhibits 100% efficiency; in practice, efficiencies of 70% to 95% are typically obtained. This is achieved using *switched-mode*, or *chopper*, circuits whose elements dissipate negligible power. *Pulse-width modulation* (PWM) allows control and regulation of the total output voltage. This approach is also employed in applications involving alternating current, including high-efficiency dc-ac power converters (inverters and power amplifiers), ac-ac power converters, and some ac-dc power converters (low-harmonic rectifiers). A basic dc-dc converter circuit known as the *buck converter*.

PROCEDURE:-

A. For Positive Output

I. Without Load.

- Make sure that power supply must be OFF.
- Refer block diagram & carry out the following connections.
- Connect power supply in proper polarity to kit BEL-PET.
- Connect M1 & M2 terminal in MOSFET bridge CKT to 4 OT winding of transformer. i.e. pin no.11 & 12 resp.
- Now select 1 OT winding of transformer and connect pin no.1 to GND.
- Next pin no 2 is connected to the anode of diode 1N5819 from diode bank.
- Cathode of same diode is connected to the terminal of 10uF/63V capacitor.
- Connect -ve terminal of capacitor to the GND.

- Observe the Amplitude of AC square wave at 4 OT winding & 2 OT winding.
- Put pot 6 at MAX position for without load condition & observe the O/P voltage & compare with load value.
- Observe the DC O/P at the terminal of capacitor .we get DC value is less than 12V DC which is used as input in MOSFET Bridge.
- Hence we obtain DC to DC conversion using Dual conversion method.

II. With Load.

- Repeat the same procedure for With Load experiment, just connect resistive 560E/2W load across the capacitor.
- Observe the O/P across the load, you will get the ripples at the O/P.
- If the voltage is less, then vary the pot 6 of PWM section to change the duty cycle. It will compensate the difference in the load value.& the actual value when load is connected.

B. For Negative Output

I. Without Load

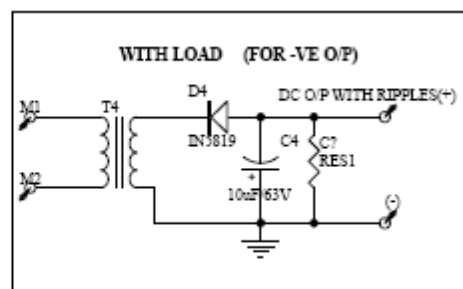
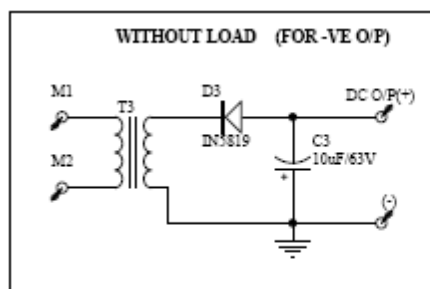
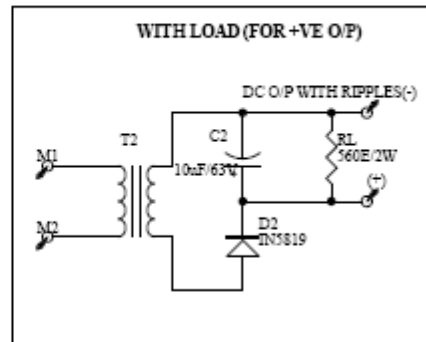
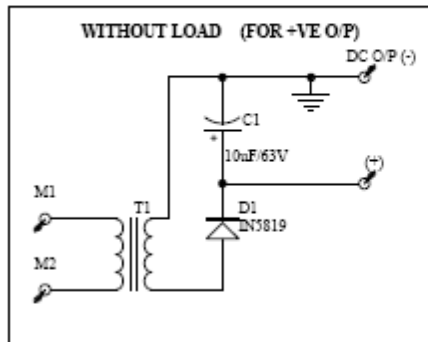
- Repeat the steps (1) to (4) from previous expt.
- Select 1OT winding of transformer & connect pin 2 to GND.
- Next pin no 1 to be connected to the cathode of diode 1N5819 from diode bank.
- Anode of same diode is connected to the -ve terminal of 10uF/63V capacitor & +ve terminal to the GND.
- Observe DC –ve voltage at –ve terminal of capacitor.

II. With Load

- Repeat the same procedure for with load condition, just connect resistive load 560E/2W load across the capacitor.
- Observe the O/P across the load; you will get ripples at the O/P.

CIRCUIT DIAGRAM

DC TO DC CONVERTER (DUAL CONVERTER)

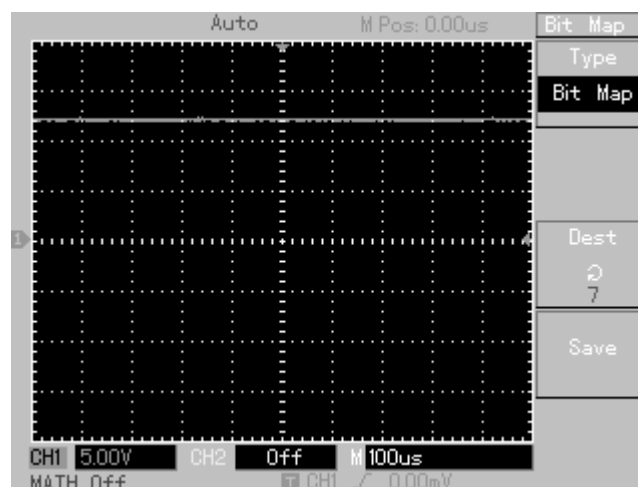


OBSERVATION:-

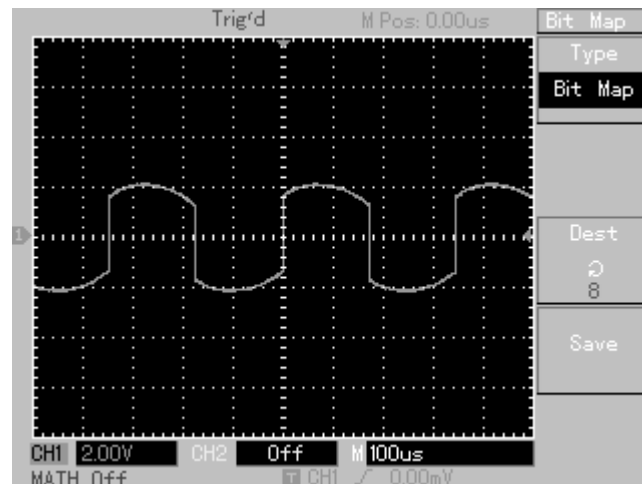
Captured Waveform

A. Without Load

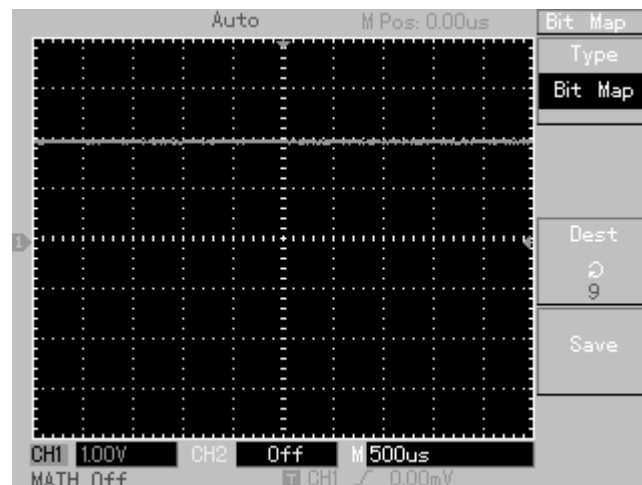
- DC Input (+12V)



- AC Signal at 10T of transformer

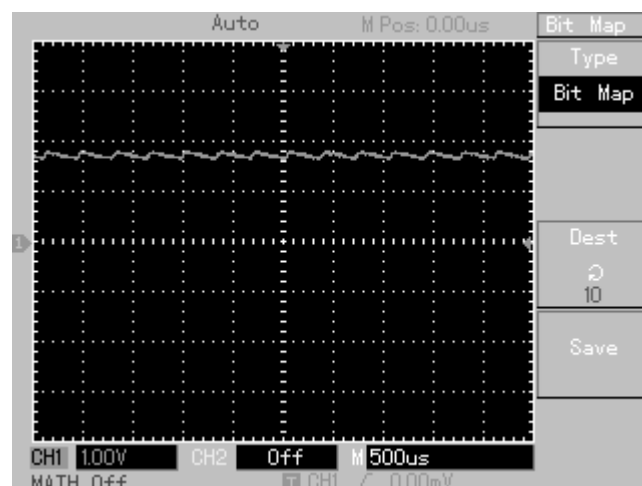


- DC output across capacitor (~2VDC)



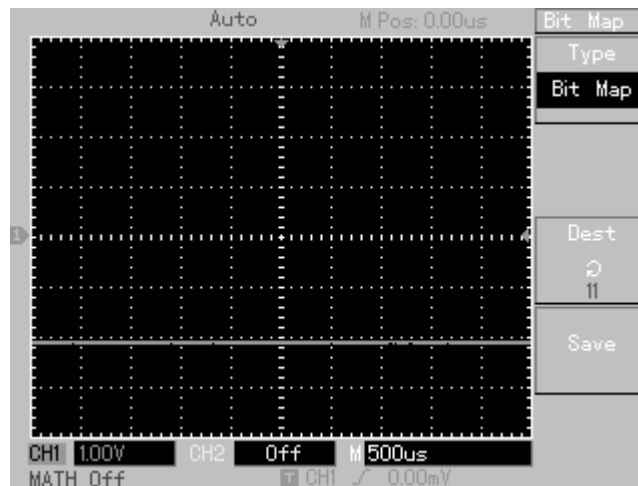
B. With Load(560ohm/2W)

- DC output across Resistance (load) with ripples.

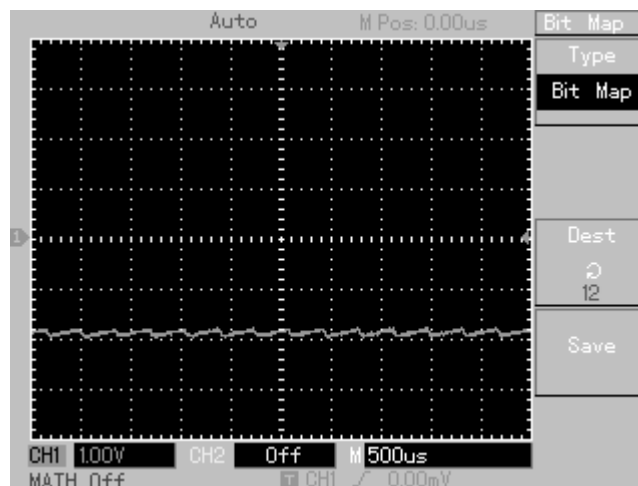


A. Without Load

- DC input & AC signal is same as that of experiment NO.2.
- DC output across capacitor (~2V DC).

**C. With Load (560OHM/2w)**

- DC output across load with ripples.



**EXPERIMENT
NO. 3**

EXPERIMENT NO: 3

NAME:-

Low Voltage Lamp Flasher (Using PUT & SCR)

OBJECTIVE:-

Study of low voltage lamp flasher (using PUT & SCR)

THEORY:-

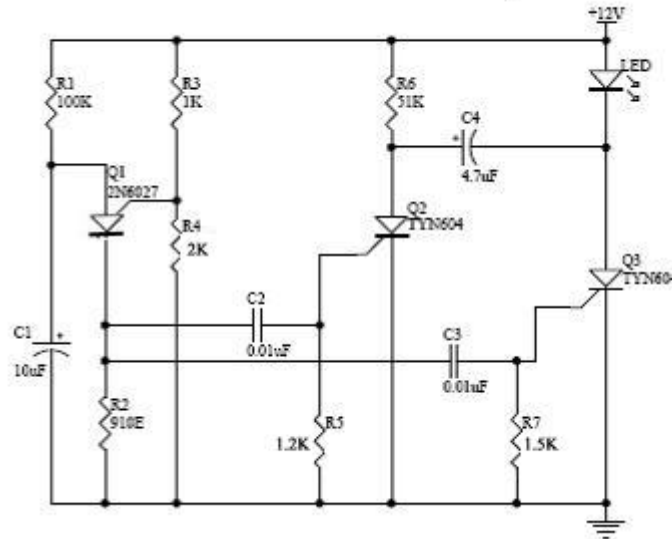
A "two-wire" circuit and simply connects in series with the load and battery. The two resistors on the base of the PNP set a threshold voltage and when power is applied the capacitor begins charging toward this voltage. When the capacitor voltage is high enough the two transistors begin to conduct. The current flow causes the voltage across the circuit to drop slightly and this drop causes a drop in the threshold voltage. The lower threshold voltage causes even more current and this positive feedback causes the circuit to rapidly turn on. It stays on until the capacitor discharges at which point a reverse process causes the circuit to suddenly switch off.

PROCEDURE:-

- Refer the circuit diagram carry out the following connections.
- Connect power supply in proper polarity to the kit BEL-PET.

CIRCUIT DIAGRAM:-

LOW VOLTAGE LAMP FLASHER (USING PUT & SCR)

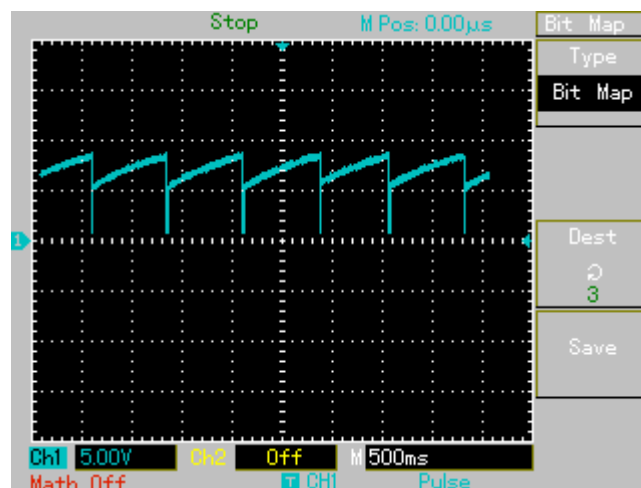


OBSERVATION:-

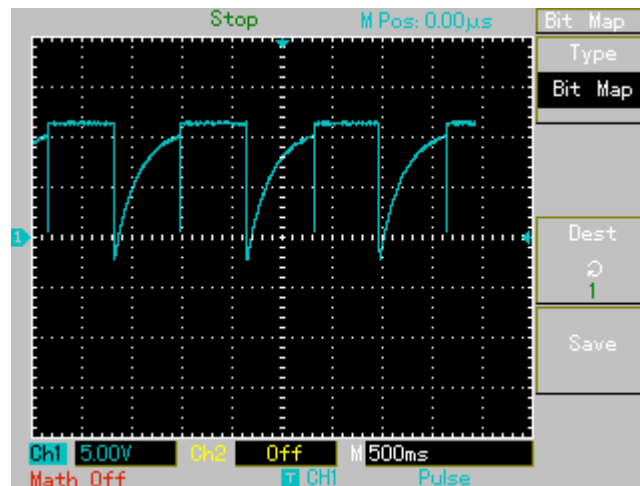
- Led will flash continuously with same ON & OFF Time.
- When we replace R5=R6=10K then LED will go ON for ½ second & OFF for same time.

CAPTURED WAVEFORM:-

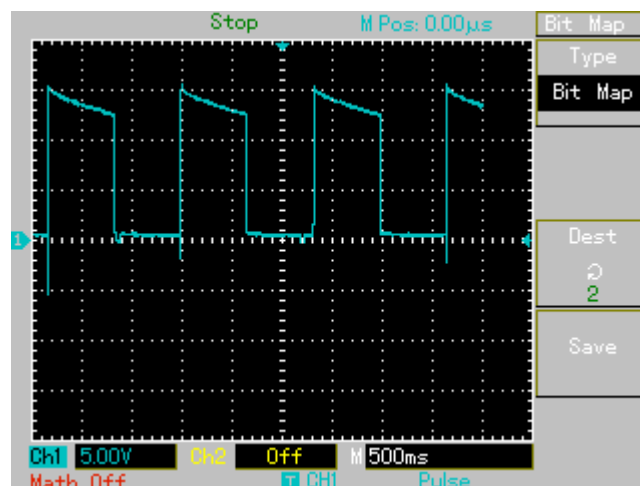
- Waveform across PUT.



- Waveform across SCR1



- Waveform across SCR2



EXPERIMENT NO. 4

EXPERIMENT NO: 4

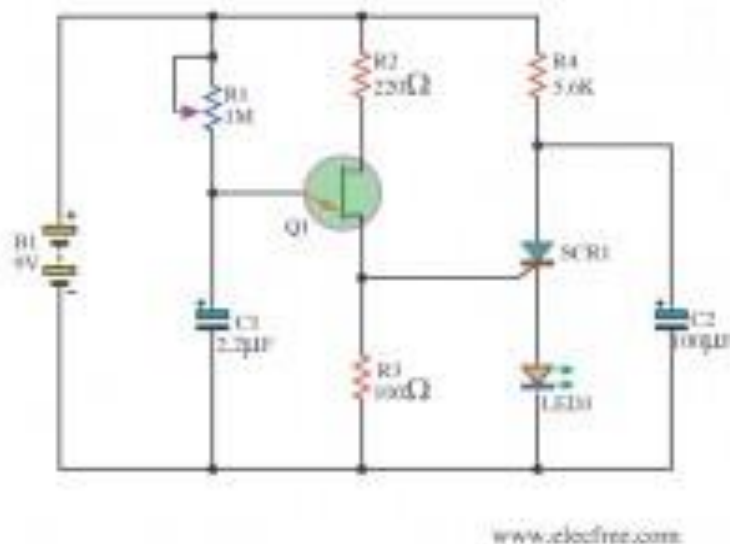
NAME:-

LED Flasher with SCR and UJT

OBJECTIVE:-

Study of LED flasher with SCR and UJT

THEORY:-



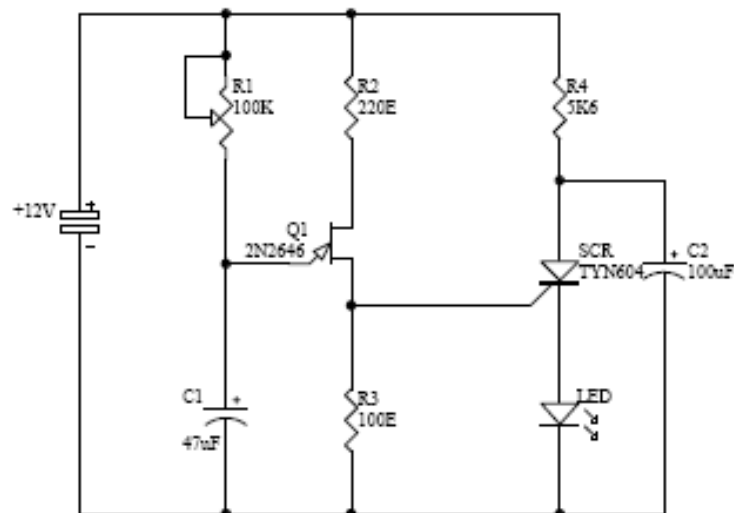
The Q1 will give the period of time of pulse for trigger. The work of SCR for drive LED. The circuit will pull current 2 mA from just Battery 9V. When the schedule is Flasher keep 12 time per second The SCR use at have 6A 50V.

LED flashers are great fun for the novice or seasoned builder. In modern times, analog circuits such as the astable BJT multivibrator or a 555 timer IC driving 4017 decade counter(s) are being replaced with small microprocessor-based circuits such as the PIC Microcontroller. The QRP/SWL Home Builder web site has featured a number of multivibrator and LED flasher circuits over the past decade and each year at least 40 - 50 emails are received requesting help, specific designs or calling for more projects. Thus, another page of LED flasher circuits was added and building these circuit was quite enjoyable.

Lower LEDs were used for all the experiments. The 2N3904 transistor was used exclusively and the schematics do not indicate this part. By request, this web page these circuit using a lot of math.

PROCEDURE:-

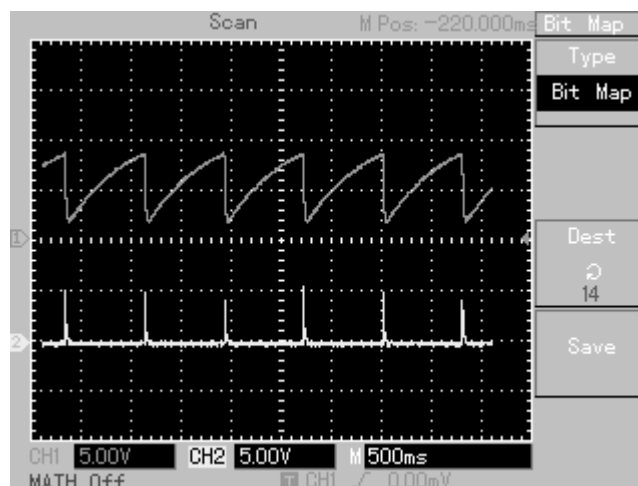
1. Refer the circuit diagram & carry out the following connection.
2. Connect power supply in proper polarity to the kit.

CIRCUIT DIAGRAM:-**LED FLASHER WITH SCR & UJT****OBSERVATION:-**

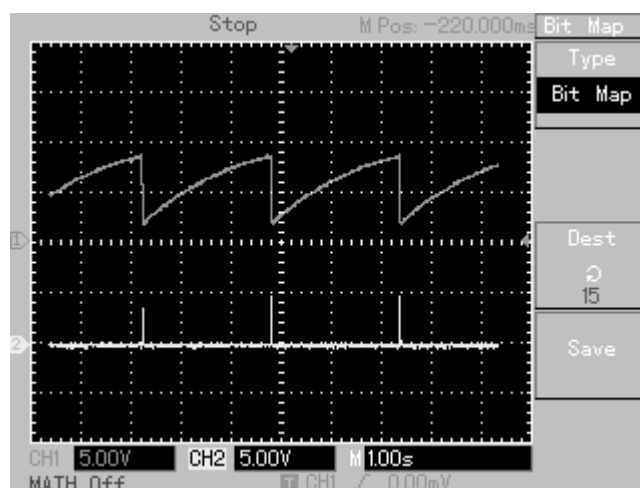
- When pot R1 is at minimum position, LED will remain ON.
- As we increasing pot R1, OFF time of LED will increase and ON time of LED is for $\frac{1}{2}$ second.

CAPTURED WAVEFORM:-

- Waveform at point E of UJT (2N2646) at R1=13K waveform across LED.



- A waveform at R1=50K.



EXPERIMENT NO. 5

EXPERIMENT NO: 5

NAME:-

Lamp Dimmer Using DIAC and TRIAC

OBJECTIVE:-

Study of lamp dimmer using diac and triac

THEORY:-

Dimmers are devices used to vary the brightness of a light. By decreasing or increasing the RMS voltage and hence the mean power to the lamp it is possible to vary the intensity of the light output. Although variable-voltage devices are used for various purposes, the term dimmer is generally reserved for those intended to control resistive incandescent, halogen and more recently compact fluorescent (CFL) lighting. More specialized pulse-width modulation equipment is needed to dim fluorescent, mercury vapor, solid state and other arc lighting. Dimmers range in size from small units the size of a normal light switch used for domestic lighting to high power units used in large theatre or architectural lighting installations. Small domestic dimmers are generally directly controlled, although remote control systems (such as X10) are available. Modern professional dimmers are generally controlled by a digital control system like DMX or DALI. In newer systems these protocols are often used in conjunction with ethernet. In the professional lighting industry changes in intensity are called “fades” and can be “fade up” or “fade down”. Dimmers with direct manual control had a limit on the speed they could be varied at but this issue has been largely eliminated with modern digital units (although very fast changes in brightness may still be avoided for other reasons like lamp life).

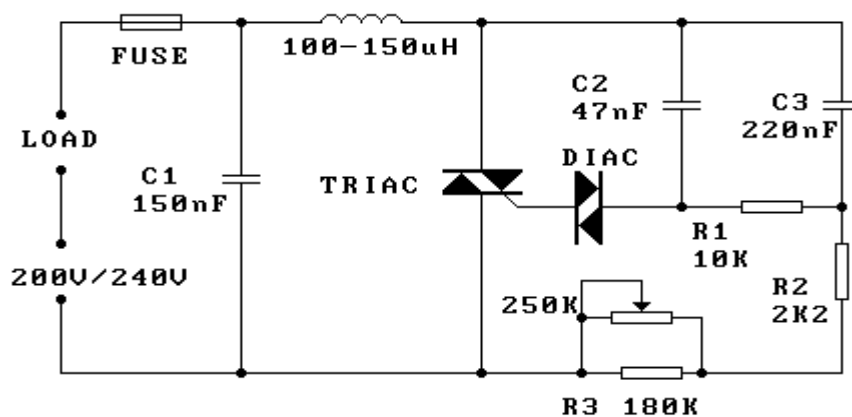


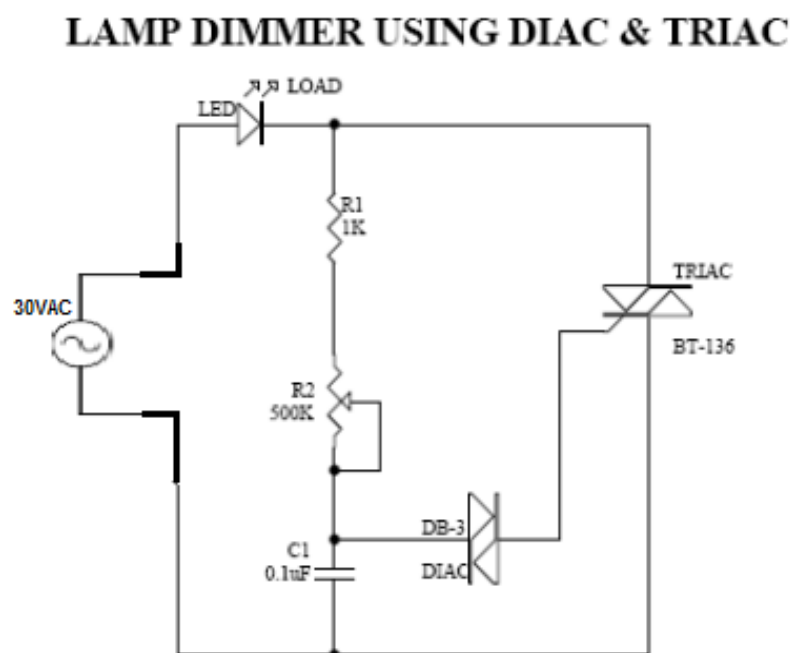
FIG:DIMMER USING DIAC AND TRIAC.

A triac may be considered as two SCR's (Silicon Controlled Rectifiers) connected in opposite directions. A diac is a gate trigger device. Triacs, diacs & SCR's are different types of Thyristors. A triac is a 3 terminal ac semiconductor switch which is triggered ON when a low energy signal is applied to its Gate. Switching is fast. The low energy of switching means that a wide range of low cost control circuits can be used, for example, optically coupled switches. Since the triac is bilateral (2 SCR's connected in opposite directions) the terms anode and cathode have no meaning. So the terms Main Terminal 1 and 2 (MT1, MT2) are used. It is standard to use MT1 as a reference point. The circuit here controls the average power to a load through the triac by phase control. The ac supply is applied to the load for only a controlled fraction of each cycle. The triac is held in an OFF condition for a portion of its cycle then is triggered ON at a time determined by the circuit. The main problem with this circuit is radiofrequency interference (RFI). Each time the triac is turned on the load current changes very quickly - a few micro seconds - from zero to a value determined by the lamp resistance and the value of the mains voltage at that instant in time. This transition generates RFI. It is greatest when the triac is triggered at 90° and least when it is triggered at close to zero or 180° of the mains AC waveform. Since there may be long lengths of mains wire between the triac and the lamp load which will radiate this RFI an L-C RFI suppression network is usually built into these types of circuits. You may detect this RFI by bringing an FM radio close to the dimmer circuit.

PROCEDURE:-

1. Connect the circuit as per shown in the diagram.
2. Now vary the pot 500K, intensity of LED will vary.

CIRCUIT DIAGRAM:-



OBSERVATION:-

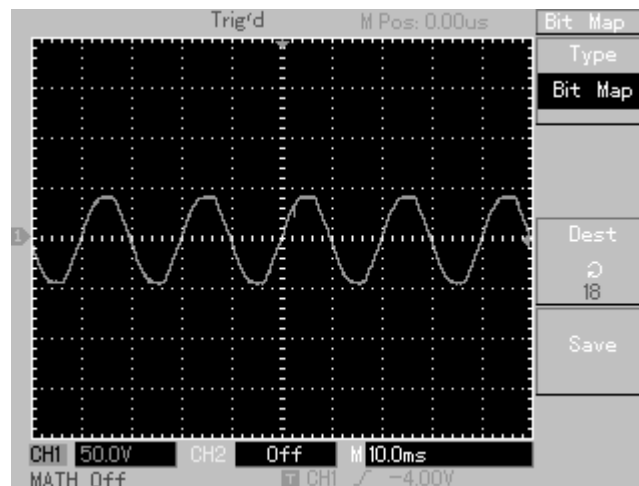
- Put pot 500K at minimum position, LED gives maximum intensity.
- As we increasing pot 500K, then intensity of LED will goes on decreasing and LED will OFF at maximum position.
- Keep CRO probe at 10X position.

Pot minimum → LED drops the current of 15 mA = MAX intensity.

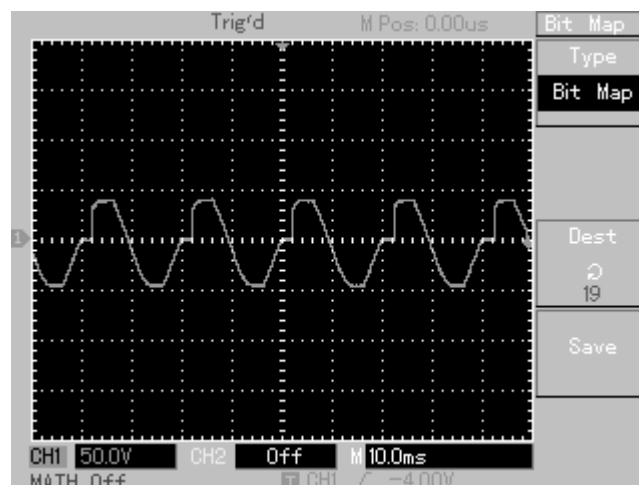
Pot at 100K → LED drops the current of 5 mA = MIN intensity.

CAPTURED WAVEFORM:-

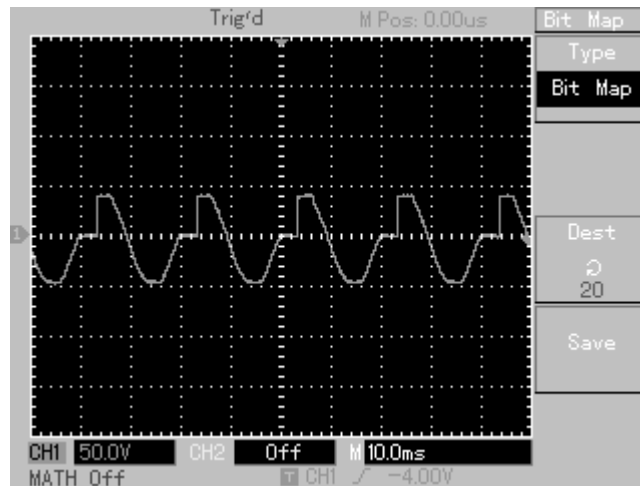
- Input AC waveform (across supply) probe at 10X.



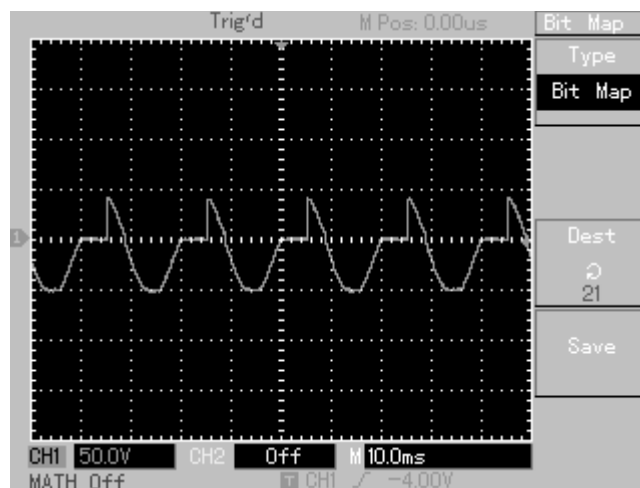
- Output across LED (POT at minimum position).



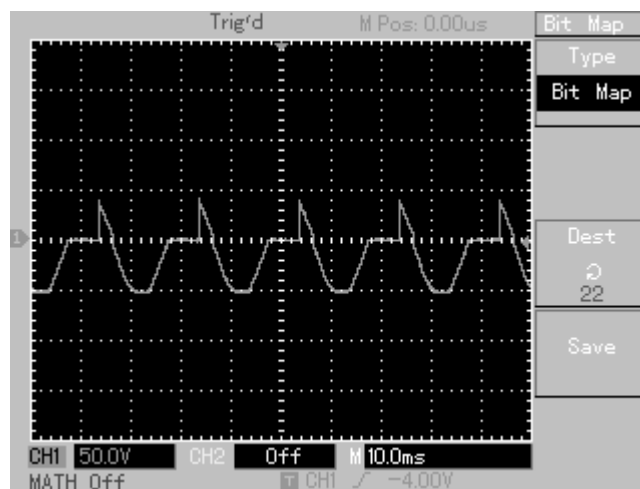
- Output across LED (POT at 40K).



- Output across LED (POT at 100K).



- Output across LED (POT at 120K).



EXPERIMENT NO. 6

EXPERIMENT NO:6

NAME:-

Half Wave Control Rectifier

OBJECTIVE:-

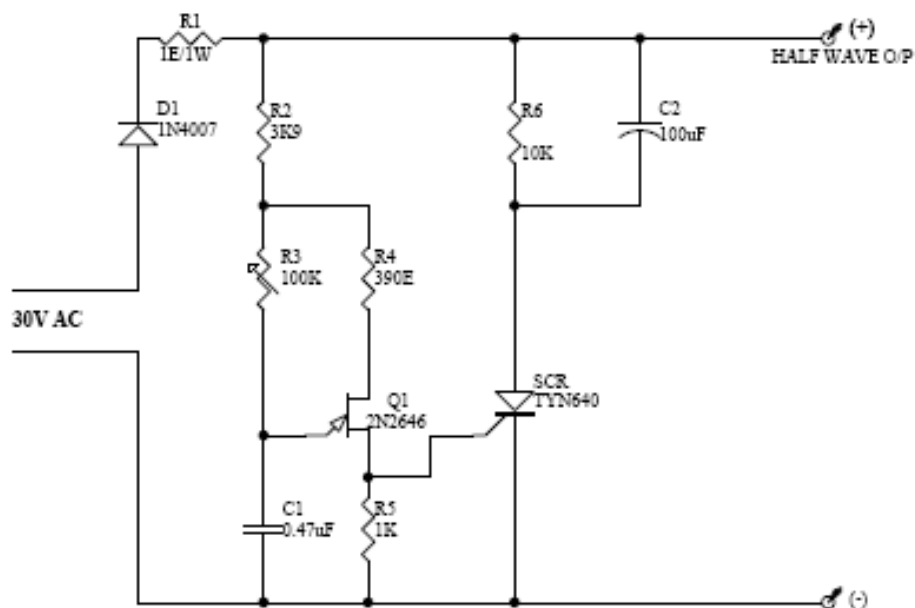
Study of half wave control rectifier

THEORY:-

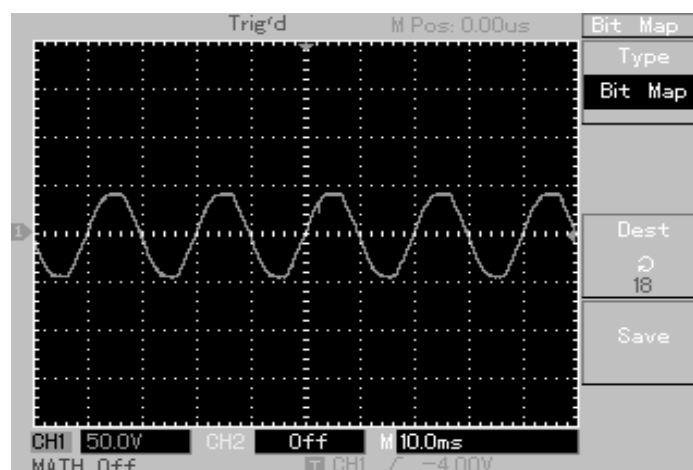
While half-wave and suffice to deliver a form of DC output, neither produces constant-voltage DC. In order to produce steady DC from a rectified AC supply, a smoothing circuit or filter is required. In its simplest form this can be just a reservoir capacitor or smoothing capacitor, placed at the DC output of the rectifier. There will still remain an amount of AC ripple voltage where the voltage is not completely smoothed. RC-Filter Rectifier: This circuit was designed and simulated using Multisim 8 software. Sizing of the capacitor represents a tradeoff. For a given load, a larger capacitor will reduce ripple but will cost more and will create higher peak currents in the transformer secondary and in the supply feeding it. In extreme cases where many rectifiers are loaded onto a power distribution circuit, it may prove difficult for the power distribution authority to maintain a correctly shaped sinusoidal voltage curve. For a given tolerable ripple the required capacitor size is proportional to the load current and inversely proportional to the supply frequency and the number of output peaks of the rectifier per input cycle. The load current and the supply frequency are generally outside the control of the designer of the rectifier system but the number of peaks per input cycle can be affected by the choice of rectifier design. A half-wave rectifier will only give one peak per cycle and for this and other reasons is only used in very small power supplies. A full wave rectifier achieves two peaks per cycle and this is the best that can be done with single-phase input. For three-phase inputs a three-phase bridge will give six peaks per cycle and even higher numbers of peaks can be achieved by using transformer networks placed before the rectifier to convert to a higher phase order.

PROCEDURE:-

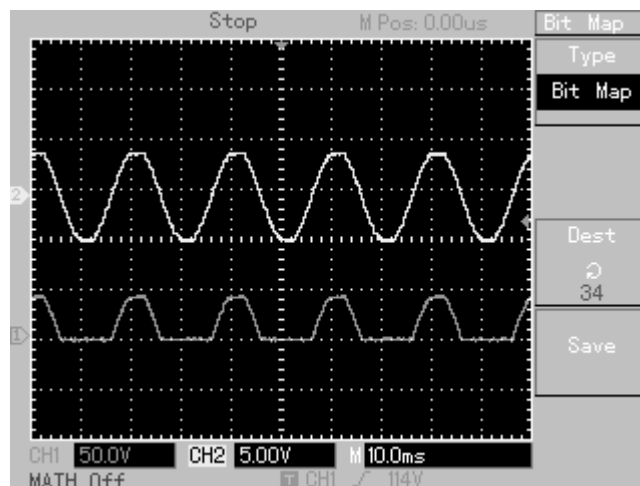
1. Refer the circuit diagram carry out the following connections.
2. Connect power supply in proper polarity to the kit BEL-PET

CIRCUIT DIAGRAM:-**HALF WAVE CONTROL RECTIFIER****OBSERVATION:-****CAPTURED WAVEFORM**

- Input AC waveform (Across supply) probe 10X.



- Half wave rectified output probe 10X.



**EXPERIMENT
NO. 7**

EXPERIMENT NO:7

NAME:-

Full Wave Control Rectifier

OBJECTIVE:-

Study of full wave control rectifier

THEORY:-

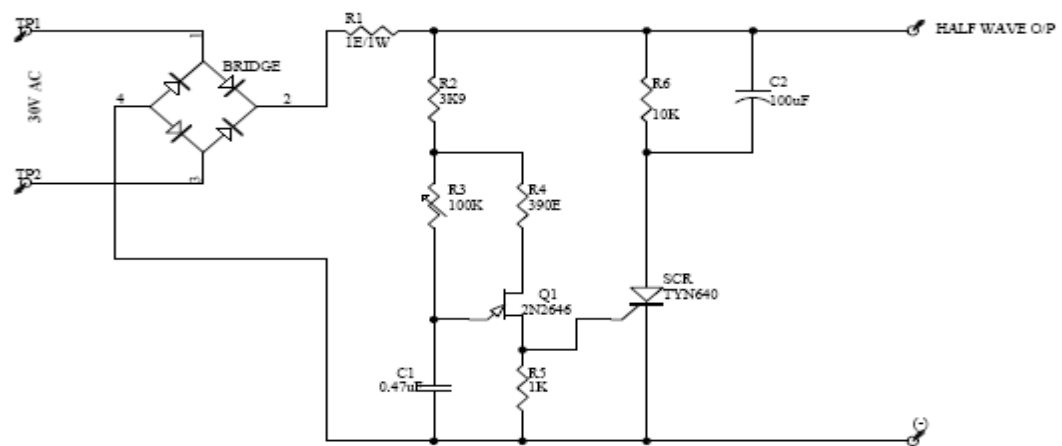
Single phase uncontrolled Full wave rectifiers suffer from poor output voltage and/or input current ripple factor. In addition, the input current contains a dc component which may cause problem (e.g. Transformer saturation etc) in the power supply system. The output dc voltage is also relatively less. Some of these problems can be addressed using a full wave rectifier. They use more number of diodes but provide higher average and rms output voltage.

There are two types of full wave uncontrolled rectifiers commonly in use. If a split power supply is available (e.g. output from a split secondary transformer) only two diode will be required to produce a full wave rectifier. These are called split secondary rectifiers and are commonly used as the input stage of a linear dc voltage regulator. However, if no split supply is available the bridge configuration of the full wave rectifier is used. This is the more commonly used full wave uncontrolled rectifier configuration.

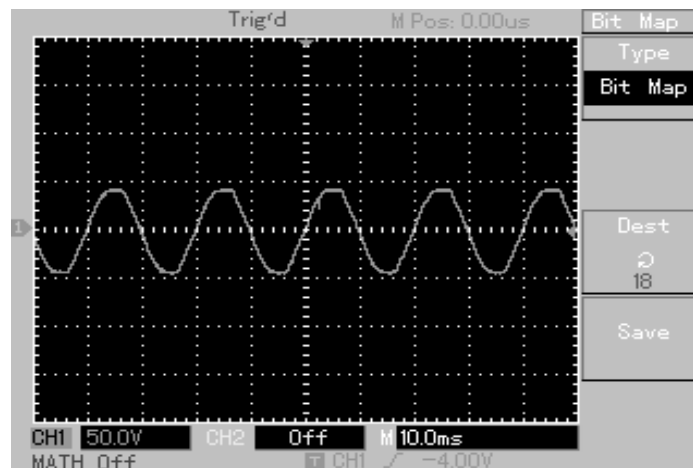
PROCEDURE:-

1. Refer the circuit diagram carry out the following connections.
2. Connect power supply in proper polarity to the kit BEL-PET

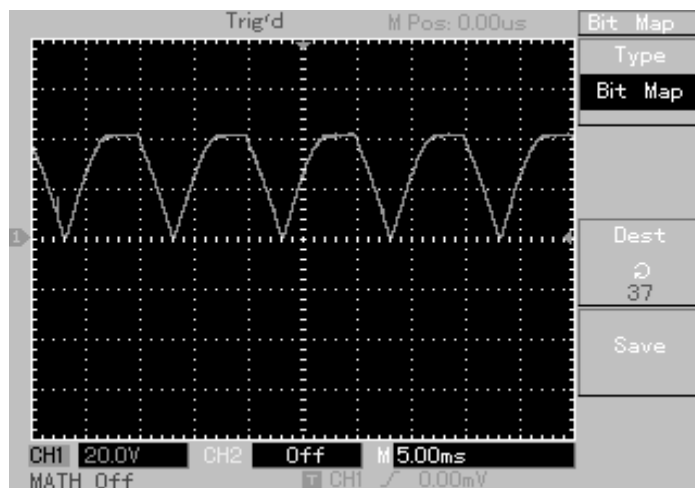
CIRCUIT DIAGRAM:-

FULL WAVE CONTROL RECTIFIER**OBSERVATION:-****CAPTURED WAVEFORM**

- AC input waveform



- Full wave rectified output



EXPERIMENT NO. 8

EXPERIMENT NO:8

NAME:-

AC Power Flasher Using SCR

OBJECTIVE:-

Study of AC power flasher using SCR

THEORY:-

A flash is a device used in photography producing a flash of artificial light (typically 1/1000 to 1/200 of a second) at a color temperature of about 5500 K to help illuminate a scene. A major purpose of a flash is to illuminate a dark scene. Other uses are capturing quickly moving objects or changing the quality of light. *Flash* refers either to the flash of light itself or to the electronic flash unit discharging the light. Most current flash units are electronic, having evolved from single-use flashbulbs and flammable powders. Modern cameras often activate flash units automatically. Flash units are commonly built directly into a camera. Some cameras allow separate flash units to be mounted via a standardized "accessory mount" bracket (hot shoe). In professional studio equipment, flashes may be large, standalone units, or *studio strobes*, powered by special battery packs or connected to mains power. They are either synchronized with the camera using a flash synchronization cable or radio signal, or are light-triggered, meaning that only one flash unit needs to be synchronized with the camera, and in turn triggers the other units

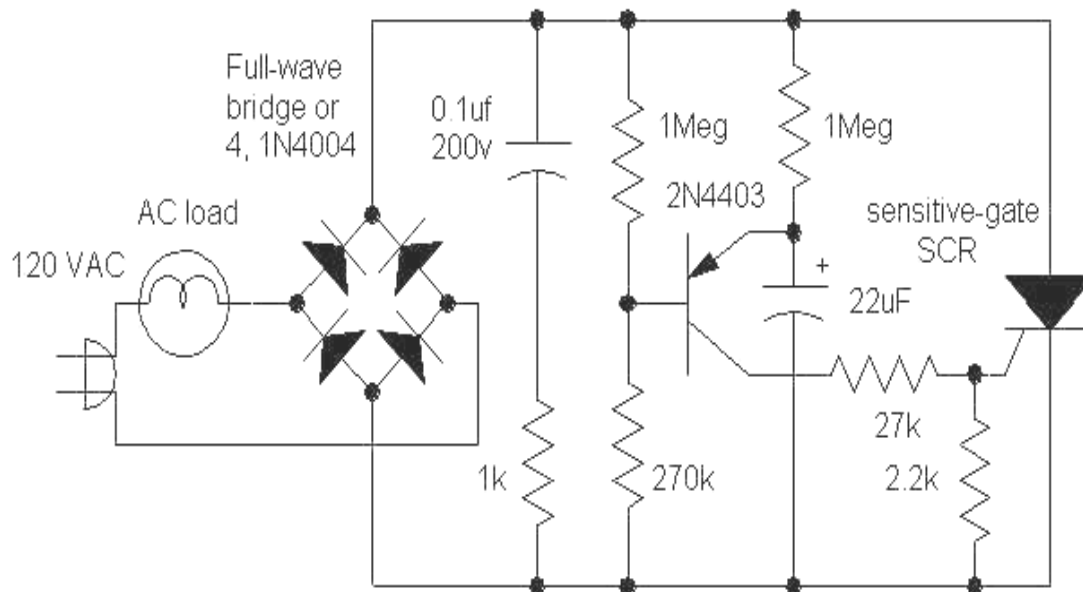


FIG:LINE POWER FLASHER

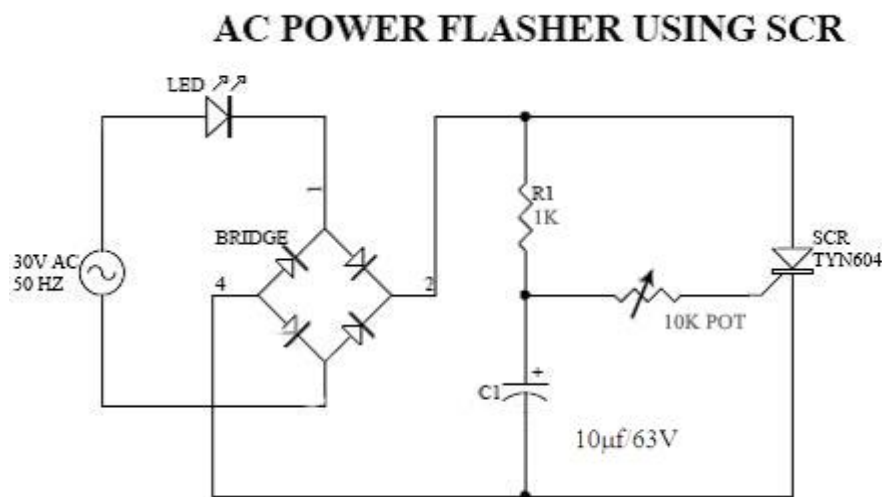
Here is an unusual flasher circuit for 120VAC loads. The circuit is similar to the two-transistor flasher seen in several circuits in techlib.com except that an SCR is used. A

little circuit trick is hiding in the selection of the 0.1 μ F capacitor in series with the 1k resistor. When the SCR is off, this capacitor smoothes the ripple from the bridge sufficiently for the transistor flasher circuitry to work properly but when the SCR turns on, the capacitor immediately discharges and will not provide enough current to keep the SCR on. Also, the capacity is low enough to leave quite a bit of ripple voltage causing the circuit to trigger near zero volts - a desirable feature

PROCEDURE:-

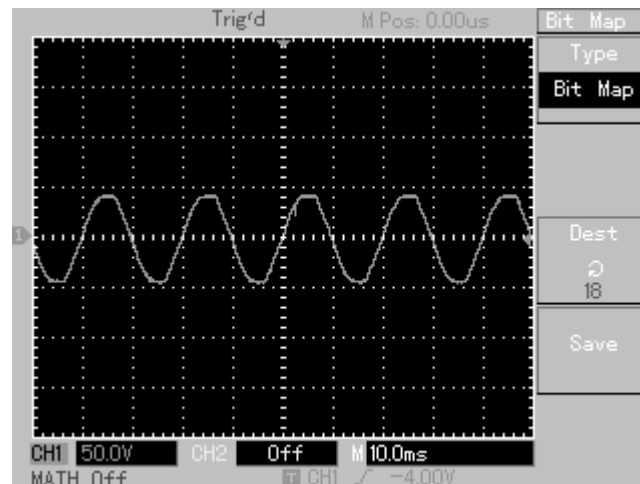
1. Refer the circuit diagram carry out the following connections.
2. Connect power supply in proper polarity to the kit BEL-PET

CIRCUIT DIAGRAM:-

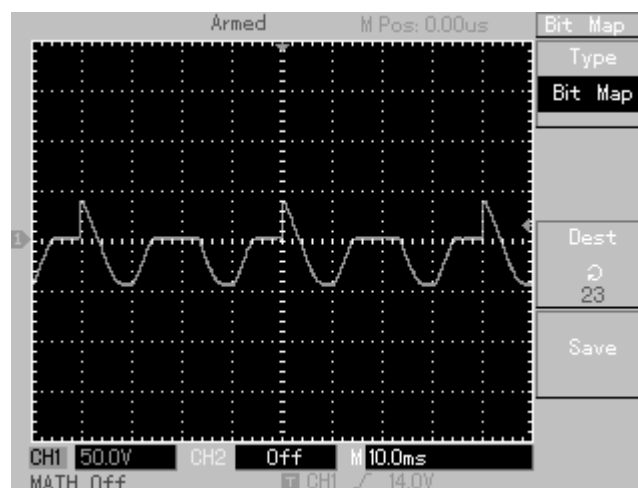


OBSERVATION:-**CAPTURED WAVEFORM**

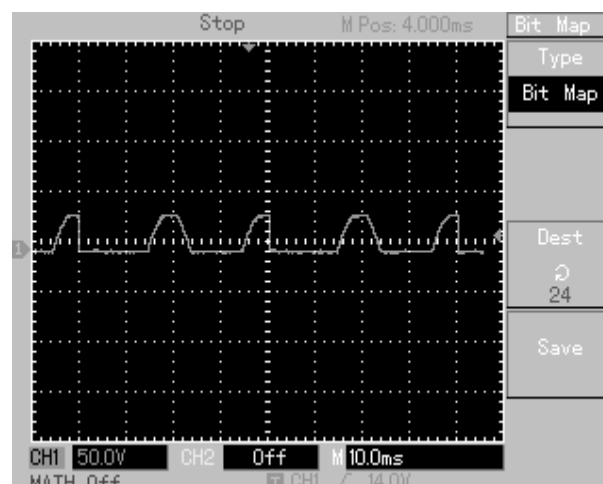
- Input AC waveform(Across supply) probe 10X



- Output across LED



- Waveform across SCR (A-K)



**EXPERIMENT
NO. 9**

EXPERIMENT NO:9

NAME:-

SCR Triggering Using 555

OBJECTIVE:-

To study SCR triggering using 555

THEORY:-

SCR triggering in the normal "off" state, the device restricts current to the leakage current. When the gate-to-cathode voltage exceeds a certain threshold, the device turns "on" and conducts current. The device will remain in the "on" state even after gate current is removed so long as current through the device remains above the holding current. Once current falls below the holding current for an appropriate period of time, the device will switch "off". If the gate is pulsed and the current through the device is below the holding current, the device will remain in the "off" state. If the applied voltage increases rapidly enough, capacitive coupling may induce enough charge into the gate to trigger the device into the "on" state; this is referred to as "dv/dt triggering." This is usually prevented by limiting the rate of voltage rise across the device, perhaps by using a snubber. "dv/dt triggering" may not switch the SCR into full conduction rapidly and the partially-triggered SCR may dissipate more power than is usual, possibly harming the device. SCRs can also be triggered by increasing the forward voltage beyond their rated breakdown voltage (also called as break over voltage), but again, this does not rapidly switch the entire device into conduction and so may be harmful so this mode of operation is also usually avoided. Also, the actual breakdown voltage may be substantially higher than the rated breakdown voltage, so the exact trigger point will vary from device to device. This device is generally used in switching applications.

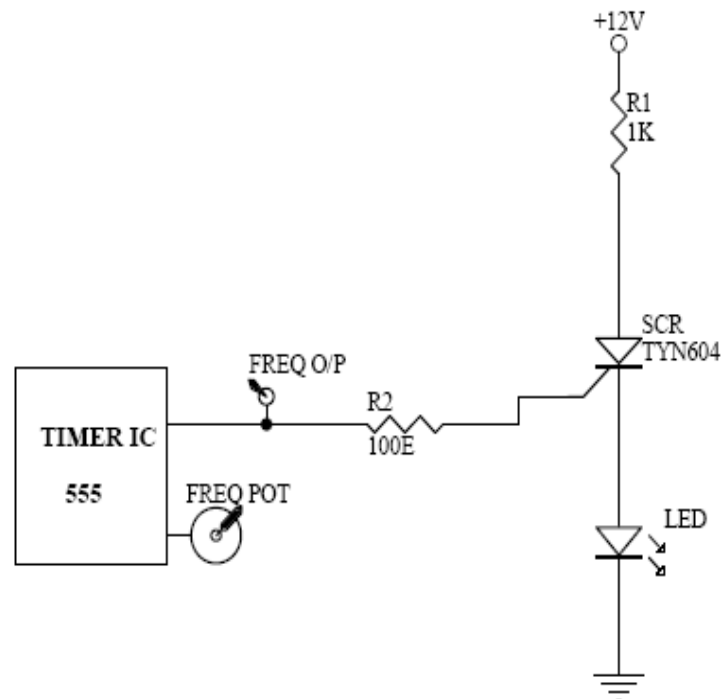
PROCEDURE:-

1. Make the Connection as per circuit diagram.
2. Keep frequency pot of 555 at MAX (1.7 KHZ) position.

OBSERVATION:-

- Observe LED flashing at O/P.
- Similarly by Varying Frequency pot of timer 555. Observe its effect on LED O/P, it will increase frequency.
- We can observe LED flashing from 0 KHZ to 35 KHZ Freq.

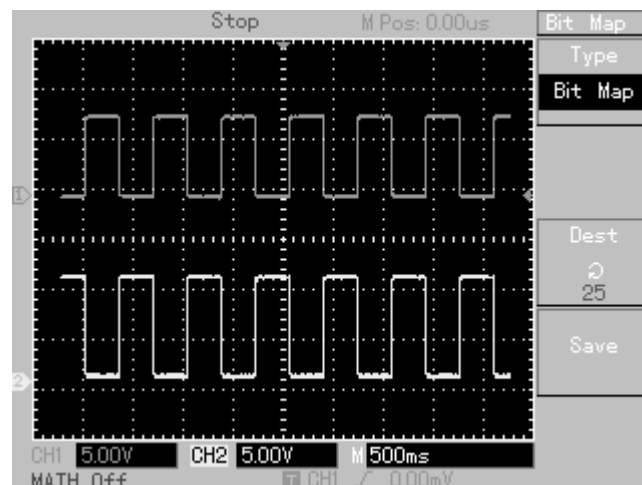
CIRCUIT DIAGRAM:-



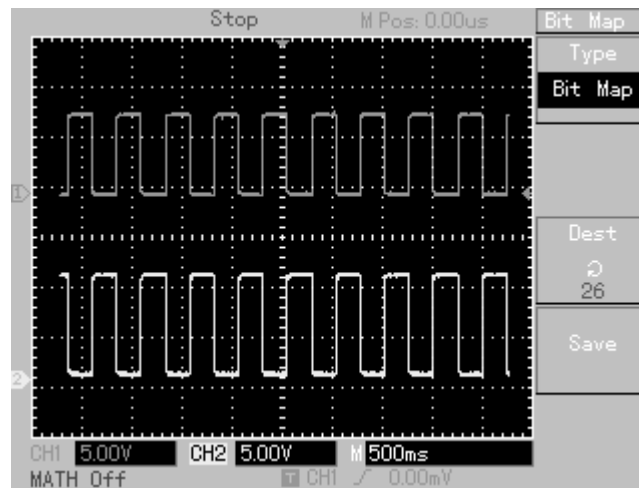
OBSERVATION:-

CAPTURED WAVEFORM

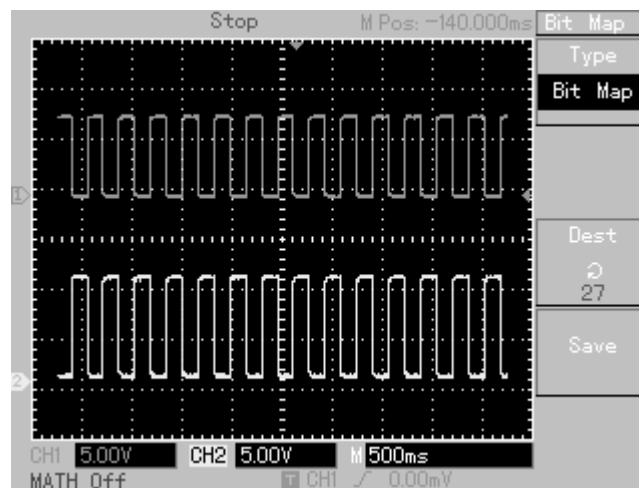
- 555 o/p pulses (POT 5 maximum i.e. Frequency =1.45HZ) waveform across SCR.



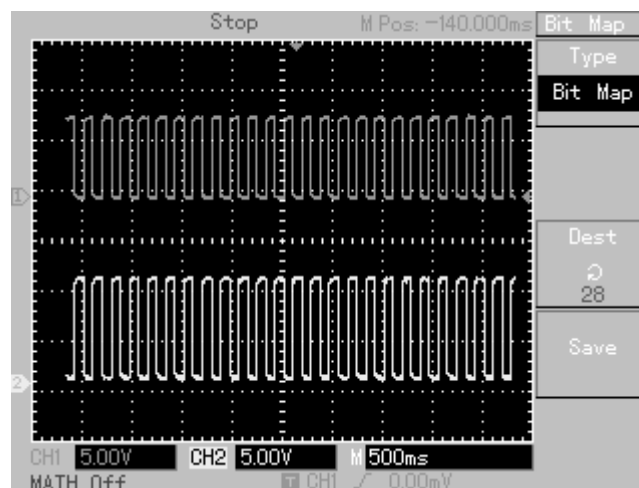
- Waveform at F=2HZ



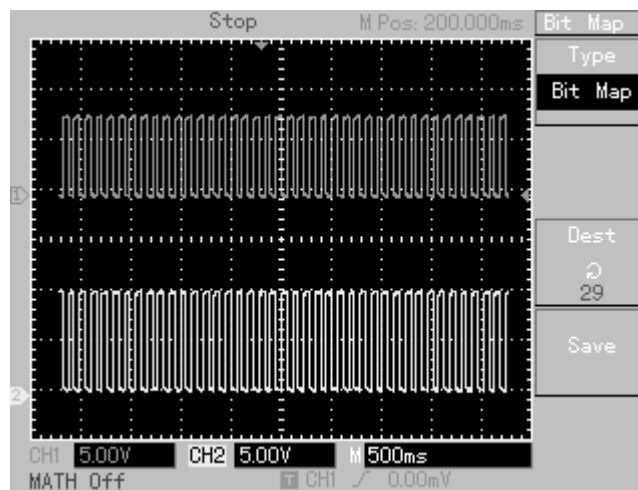
- Waveform at $F=3\text{Hz}$.



- Waveform at $F=5\text{Hz}$.



- Waveform at $F=9\text{Hz}$.



**EXPERIMENT
NO. 10**

EXPERIMENT NO:10

NAME:-

SCR Time Delay Relay

OBJECTIVE:-

To study SCR time delay relay

THEORY:-

Time delay relays are built in these four basic modes of contact operation:

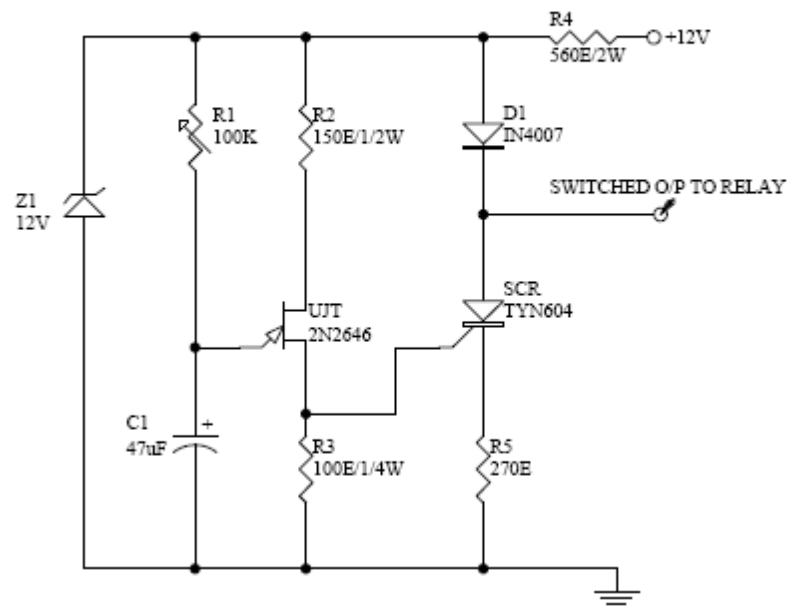
- Normally-open, timed-closed. Abbreviated "NOTC", these relays open immediately upon coil de-energization and close only if the coil is continuously energized for the time duration period. Also called normally-open, on-delay relays.
- Normally-open, timed-open. Abbreviated "NOTO", these relays close immediately upon coil energization and open after the coil has been de-energized for the time duration period. Also called normally-open, off delay relays
- Normally-closed, timed-open. Abbreviated "NCTO", these relays close immediately upon coil de-energization and open only if the coil is continuously energized for the time duration period. Also called normally-closed, on-delay relays.
- Normally-closed, timed-closed. Abbreviated "NCTC", these relays open immediately upon coil energization and close after the coil has been de-energized for the time duration period. Also called normally-closed, off delay relays. One-shot timers provide a single contact pulse of specified duration for each coil energization (transition from coil off to coil on). Recycle timers provide a repeating sequence of on-off contact pulses as long as the coil is maintained in an energized state. Watchdog timers actuate their contacts only if the coil fails to be continuously sequenced on and off (energized and de-energized) at a minimum frequency.

PROCEDURE:-

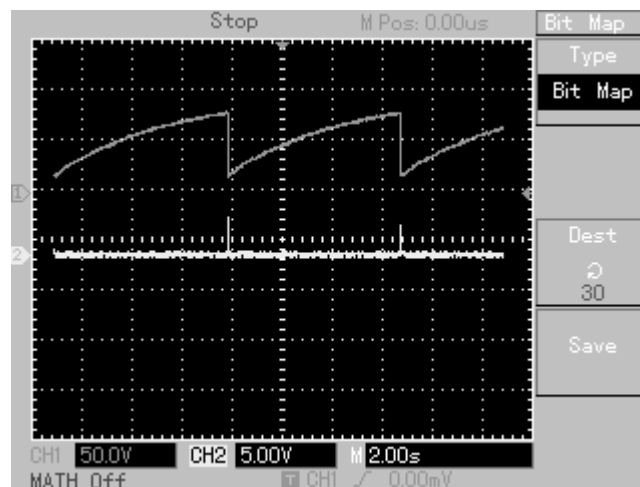
1. Connect the circuit as per circuit diagram.

OBSERVATIONS:-

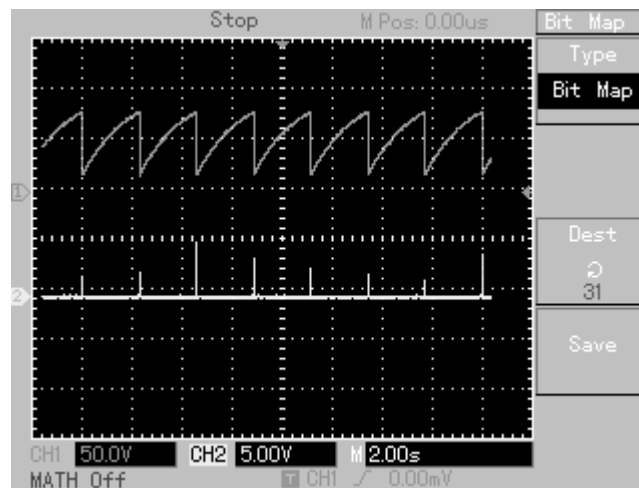
- At Min position of pot R1 we get 9.65V at output.(Relay ON)
- At Max position of pot R1 we get 0.4V at output.(Relay OFF)

CIRCUIT DIAGRAM:-**SCR TIME DELAY RELAY****OBSERVATION:-****CAPTURED WAVEFORM**

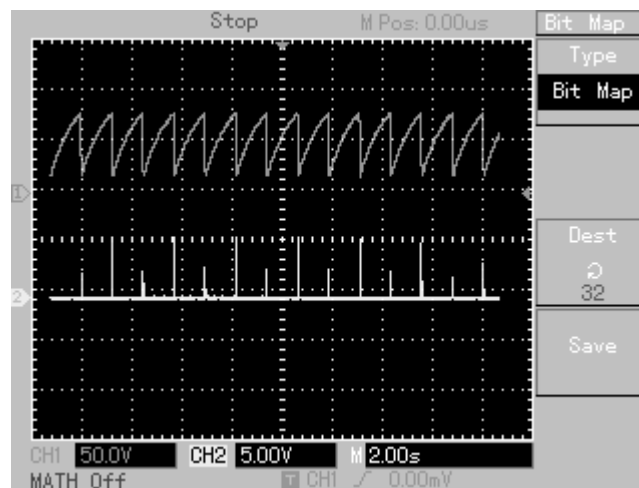
- Waveform at emitter of UJT & Wave at gate of SCR POT at maximum position.



- At POT-MID position.



- At POT-MINIMUM position.



EXPERIMENT NO. 11

EXPERIMENT NO:11

NAME:-

DC Chopper

OBJECTIVE:-

To study DC chopper

THEORY:-

In electronics, a chopper circuit is used to refer to numerous types of electronic switching devices and circuits used in power control and signal applications. A chopper is a switching device that converts fixed DC input to a variable DC output voltage directly. Essentially, a chopper is an electronic switch that is used to interrupt one signal under the control of another.

In power electronics applications, since the switching element is either fully on or fully off, its losses are low, and the circuit can provide high efficiency. However, the current supplied to the load is discontinuous and may require smoothing or a high switching frequency to avoid undesirable effects. In signal processing circuits, use of a chopper stabilizes a system against drift of electronic components; the original signal can be recovered after amplification or other processing by a synchronous demodulator that essentially un-does the "chopping" process.

For all the chopper configurations operating from a fixed DC input voltage, the average value of the output voltage is controlled by periodic opening and closing of the switches used in the chopper circuit. The average output voltage can be controlled by different techniques namely:

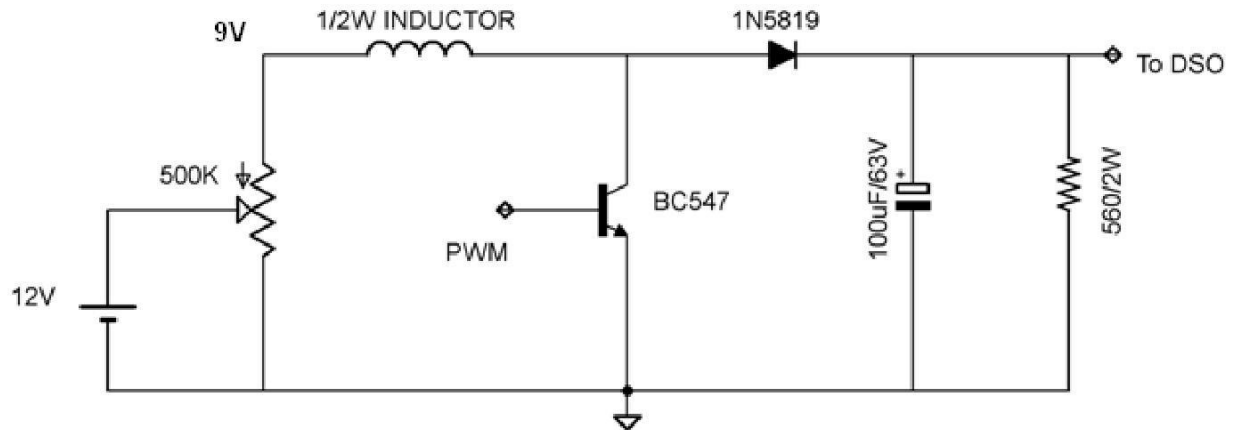
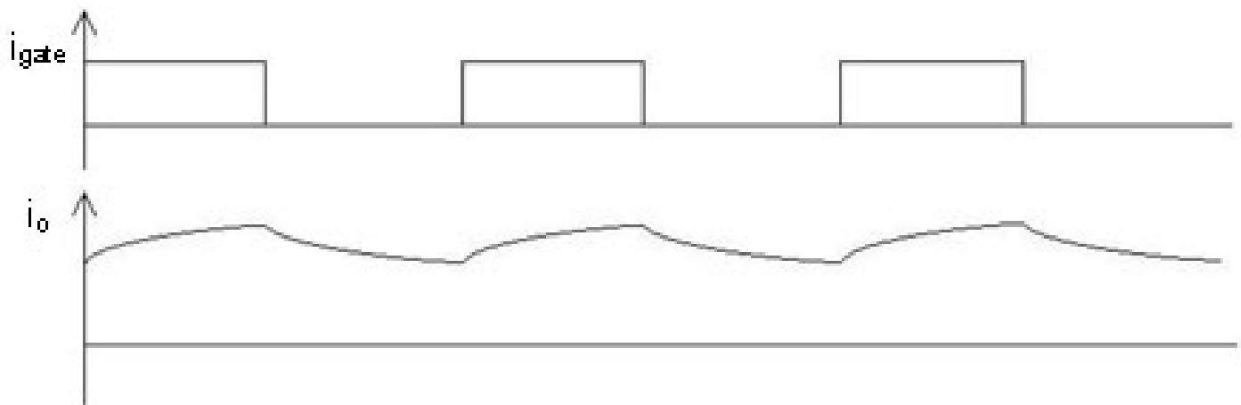
- Pulse width modulation technique
- Frequency modulation
- Variable frequency, variable pulse width
- CLC control

PROCEDURE:-

- Connect the circuit as per circuit diagram.
- First create +9V by using 500K POT and 12V supply.
- Then connect 9V to circuit as shown in fig.
- Change the ON time of PWM wave then observed output.

OBSERVATIONS:-

- Observe the waveform of PWM and note down the frequency.
- Observe the waveform across load resistor 560 Ω 2W.

CIRCUIT DIAGRAM:-**DC CHOPPER WAVEFORMS**

**EXPERIMENT
NO. 12**

EXPERIMENT NO:12

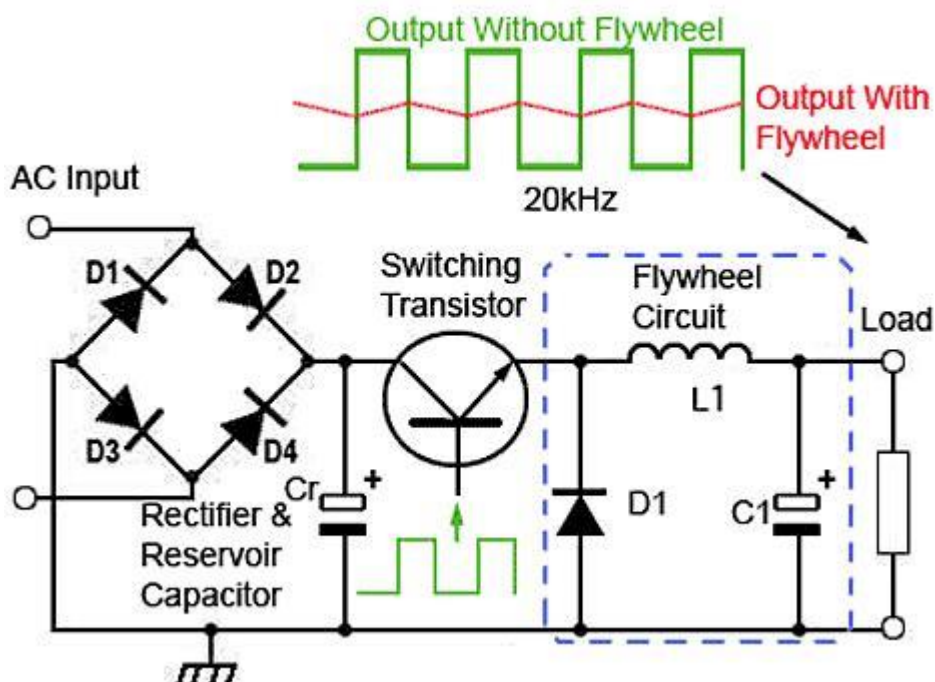
NAME:-

Switched Mode Converter / Rectifier.

OBJECTIVE:-

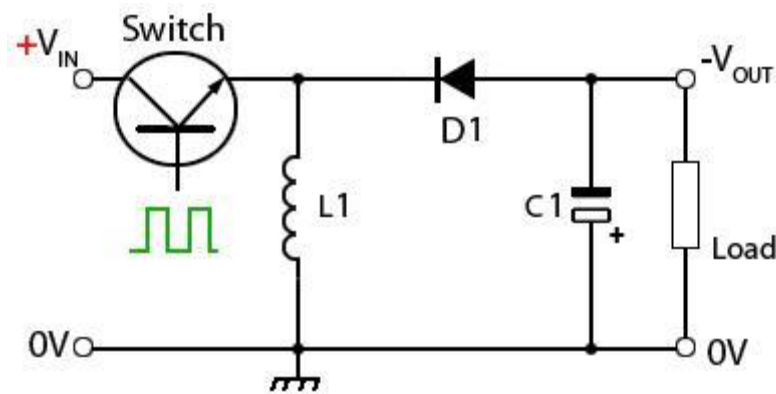
To study Switched Mode Converter / Rectifier

THEORY:-



The Switched Converter is used in SMPS circuits where the DC output voltage needs to be lower than the DC input voltage. The DC input can be derived from rectified AC or from any DC supply. It is useful where electrical isolation is not needed between the switching circuit and the output, but where the input is from a rectified AC source, isolation between the AC source and the rectifier could be provided by a mains isolating transformer.

The Switching Converter also known as Buck Converter. The switching transistor between the input and output of the Buck Converter continually switches on and off at high frequency. To maintain a continuous output, the circuit uses the energy stored in the inductor L, during the on periods of the switching transistor, to continue supplying the load during the off periods. The circuit operation depends on what is sometimes also called a Flywheel Circuit. This is because the circuit acts rather like a mechanical flywheel that, given regularly spaced pulses of energy keeps spinning smoothly (outputting energy) at a steady rate.



For negative supplies the circuit shown in above fig. can be used. This involves a change around in the positions of L1 and D1, and reversing the polarity of C compared to the circuit in Fig 3.1.2. This variation of the basic buck converter now inverts the positive DC input to produce a negative supply in the range of 0V to $-V_{IN}$.

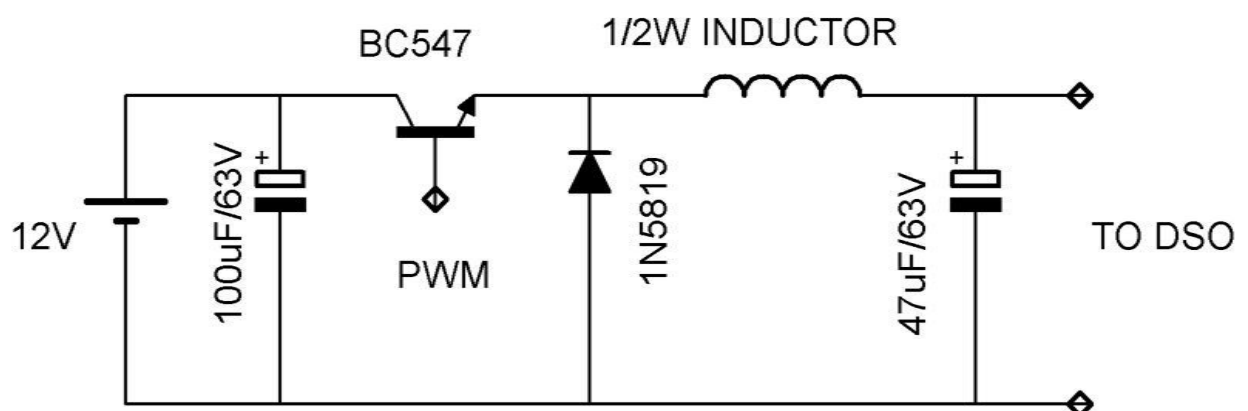
PROCEDURE:-

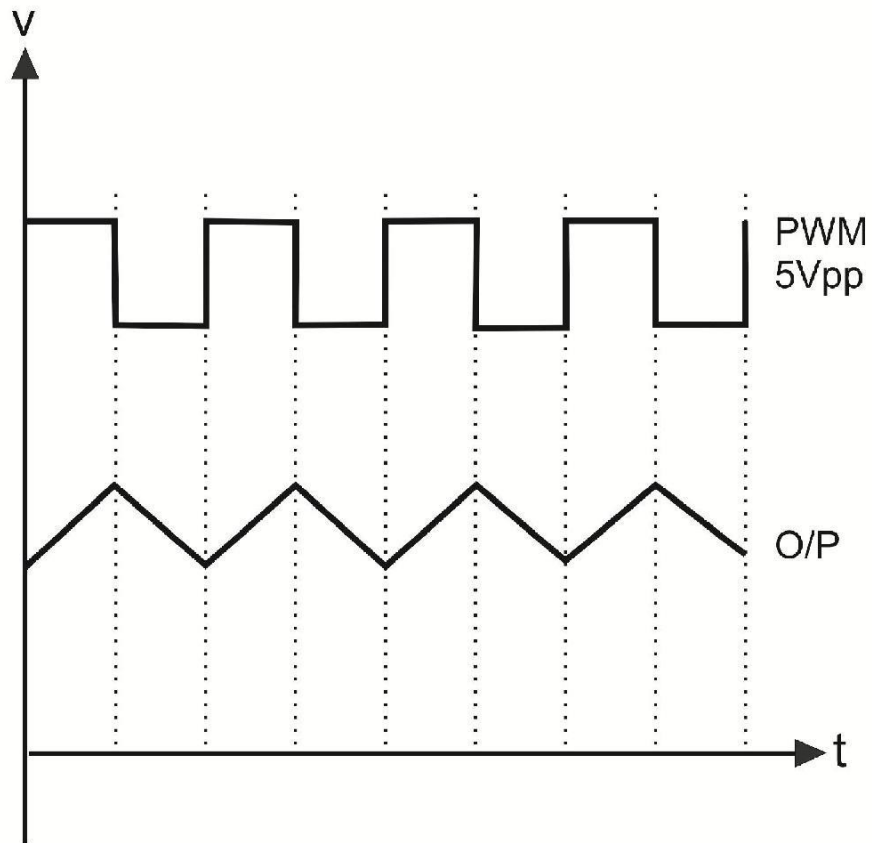
- Connect the circuit as per circuit diagram.
- Do the proper setting of DSO to observe o/p waveform.
- Use HPWM/LPWM as oscillated signal.

OBSERVATIONS:-

- Observe the waveform of PWM and note down the frequency.
- Observe the waveform across Capacitor 47 F/63V.

CIRCUIT DIAGRAM:-



WAVEFORM:-

**EXPERIMENT
NO. 13**

EXPERIMENT NO: 13

NAME:-

SCR Turn ON Methods.

OBJECTIVE:-

To study SCR Turn ON Methods.

THEORY:-

The various SCR triggering methods are

- Forward Voltage Triggering
- Thermal or Temperature Triggering
- Radiation or Light triggering.
- dv/dt Triggering
- Gate Triggering

(a) Forward Voltage Triggering:-

- In this mode, an additional forward voltage is applied between anode and cathode.
- When the anode terminal is positive with respect to cathode(V_{AK}), Junction J1 and J3 is forward biased and junction J2 is reverse biased.
- No current flows due to depletion region in J2 is reverse biased (except leakage current).
- As V_{AK} is further increased, at a voltage V_{BO} (Forward Break Over Voltage) the junction J2 undergoes avalanche breakdown and so a current flows and the device tends to turn ON(even when gate is open)

(b) Thermal (or) Temperature Triggering:-

- The width of depletion layer of SCR decreases with increase in junction temperature.
- Therefore in SCR when V_{AR} is very near its breakdown voltage, the device is triggered by increasing the junction temperature.
- By increasing the junction temperature the reverse biased junction collapses thus the device starts to conduct.

(c) Radiation Triggering (or) Light Triggering:-

- For light triggered SCRs a special terminal niche is made inside the inner P layer instead of gate terminal.

- When light is allowed to strike this terminal, free charge carriers are generated.
- When intensity of light becomes more than a normal value, the thyristor starts conducting.
- This type of SCRs are called as LASCR

(d) dv/dt Triggering:-

- When the device is forward biased, J1 and J3 are forward biased, J2 is reverse biased.
- Junction J2 behaves as a capacitor, due to the charges existing across the junction.
- If voltage across the device is V, the charge by Q and capacitance by C then,

$$i_c = dQ/dt$$

$$Q = CV$$

$$i_c = d(CV) / dt$$

$$= C \cdot dV/dt + V \cdot dC/dt$$

$$\text{as } dC/dt = 0$$

$$i_c = C \cdot dV/dt$$
- Therefore when the rate of change of voltage across the device becomes large, the device may turn ON, even if the voltage across the device is small.

(e) Gate Triggering:-

- This is most widely used SCR triggering method.
- Applying a positive voltage between gate and cathode can Turn ON a forward biased thyristor.
- When a positive voltage is applied at the gate terminal, charge carriers are injected in the inner P-layer, thereby reducing the depletion layer thickness.

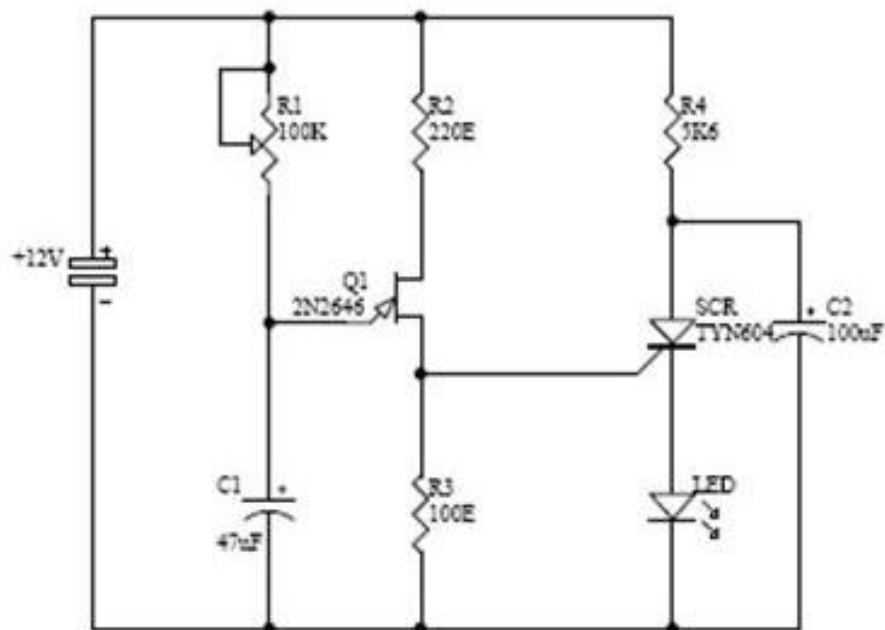
As the applied voltage increases, the carrier injection increases, therefore the voltage at which forward break-over occurs decreases.

PROCEDURE:-

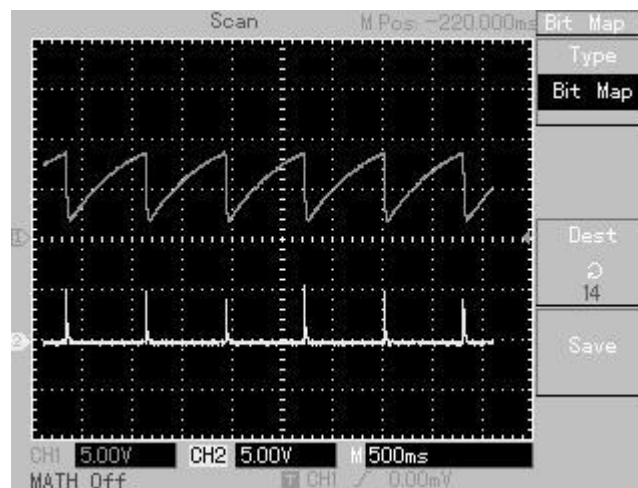
- We will perform Gate triggering of SCR using UJT.
- Connect the circuit as per circuit diagram.

OBSERVATION:-

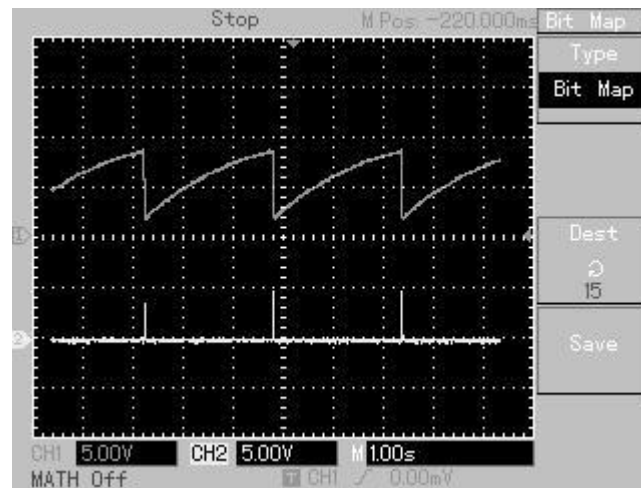
- When pot R1 is at minimum position, LED will remain ON.
- As we increasing pot R1, OFF time of LED will increase and ON time of LED is for ½ second.

CIRCUIT DIAGRAM:-**Gate Triggering Method Using UJT****CAPTURED WAVEFORM:-**

Waveform at point E of UJT (2N2646) at R1=13K waveform across LED.



A waveform at $R1=50K$.



**EXPERIMENT
NO. 14**

EXPERIMENT NO:14

NAME:-

SCR Firing circuits.

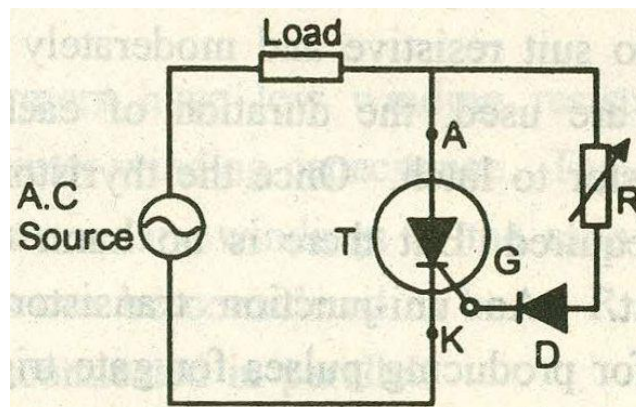
OBJECTIVE:-

To study SCR Firing circuits.

THEORY:-

(i) Resistance triggering:

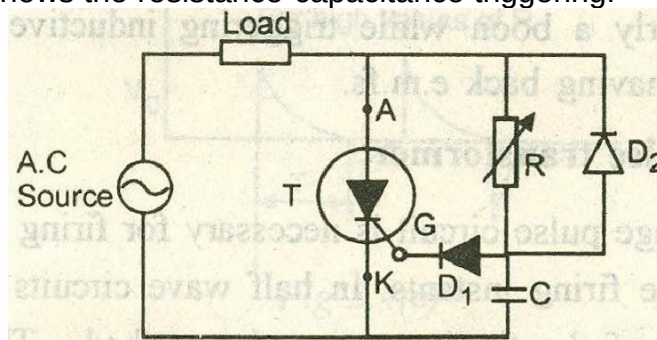
The following circuit shows the resistance triggering.



- In this method, the variable resistance R is used to control the gate current.
- Depending upon the value of R , when the magnitude of the gate current reaches the sufficient value (latching current of the device) the SCR starts to conduct.
- The diode D is called as blocking diode. It prevents the gate cathode junction from getting damaged in the negative half cycle.
- By considering that the gate circuit is purely resistive, the gate current is in phase with the applied voltage.
- By using this method we can achieve maximum firing angle up to 90°

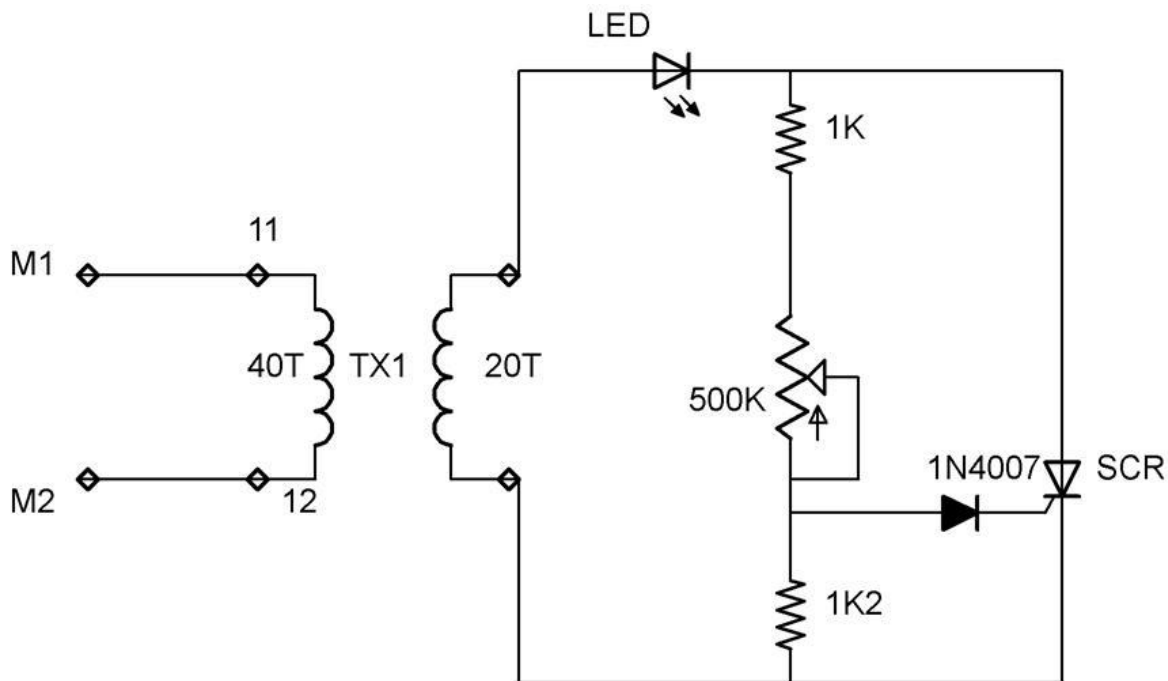
(ii) RC Triggering

The following circuit shows the resistance-capacitance triggering.



- By using this method we can achieve firing angle more than 90° .
- In the positive half cycle, the capacitor is charged through the variable resistance R up to the peak value of the applied voltage.
- The variable resistor R controls the charging time of the capacitor.
- Depends upon the voltage across the capacitor, when sufficient amount of gate current will flow in the circuit, the SCR starts to conduct.
- In the negative half cycle, the capacitor C is charged up to the negative peak value through the diode D2.
- Diode D1 is used to prevent the reverse break down of the gate cathode junction in the negative half cycle.

R FIRING CIRCUIT:



PROCEDURE:-

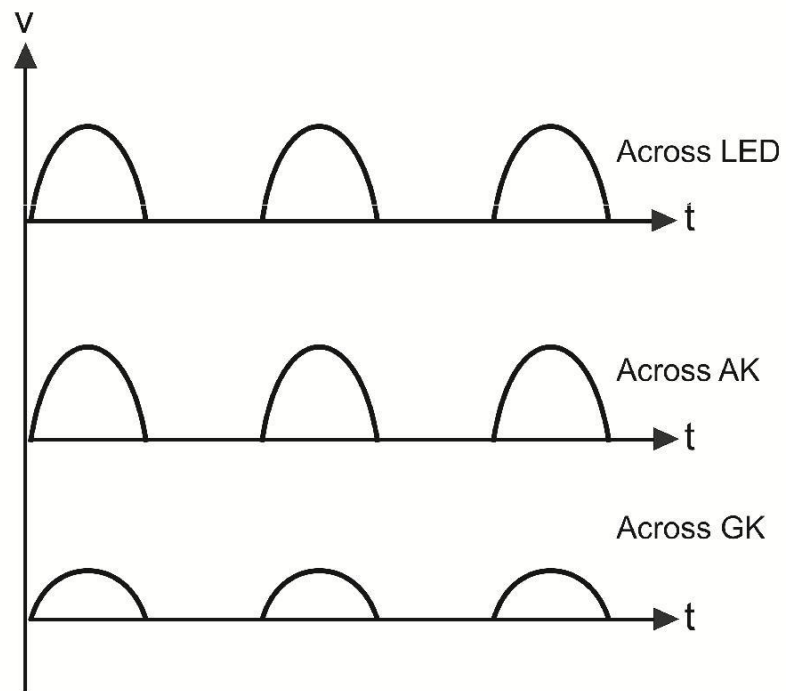
- Connect the circuit as per circuit diagram.
- Keep the PWM knob at minimum position.
- Keep the knob of 500K POT at minimum position.
- Power ON the kit.
- Slowly increase the resistance of POT. Observe the o/p.

OBSERVATION:-

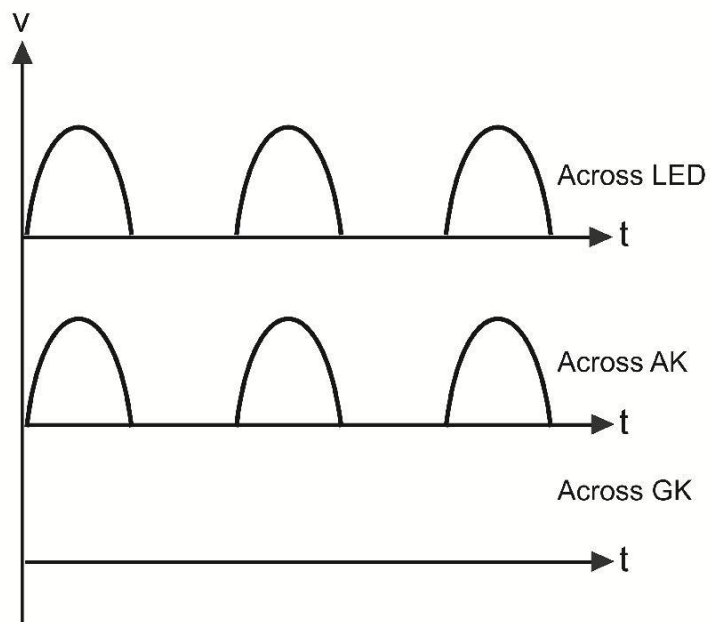
- Observe the waveform at 20T.
- Observe the waveform across LED.
- Observe the waveform across different terminals of SCR at a different position of potentiometer.

Waveform of R firing circuit of SCR:

When POT. Resistor at minimum position

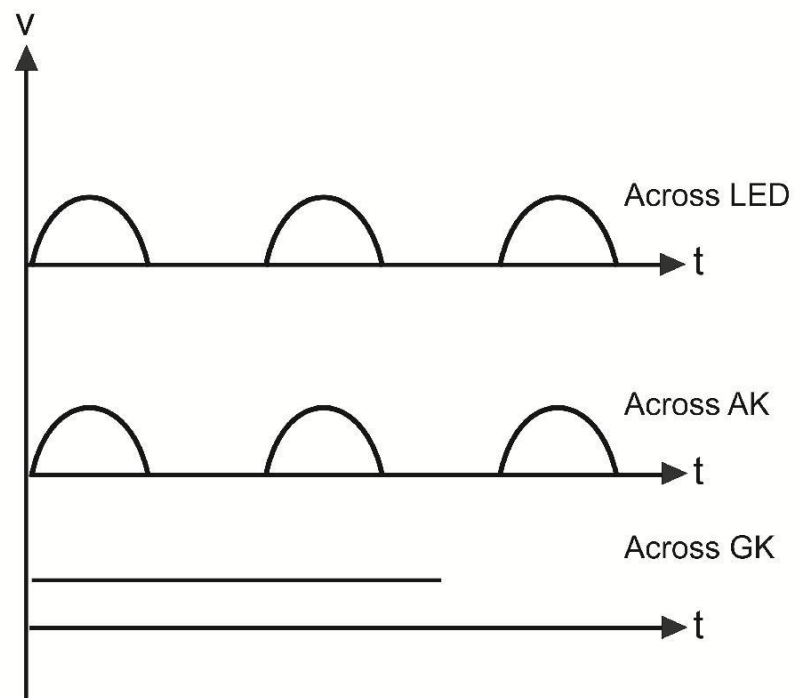


When POT Resistor at maximum position

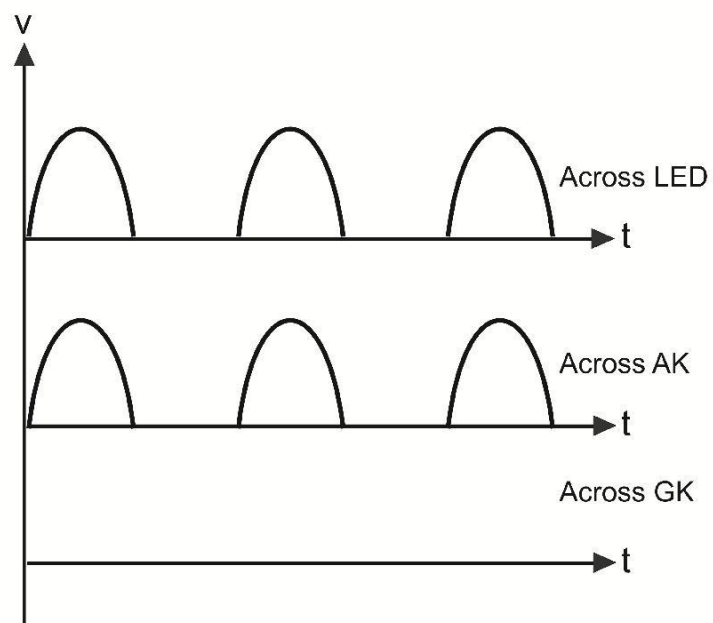


Waveform of RC firing circuit of SCR

When POT Resistor at minimum position



When POT Resistor at maximum position



EXPERIMENT NO. 15

EXPERIMENT NO:15

NAME:-

Time delay Circuit.

OBJECTIVE:-

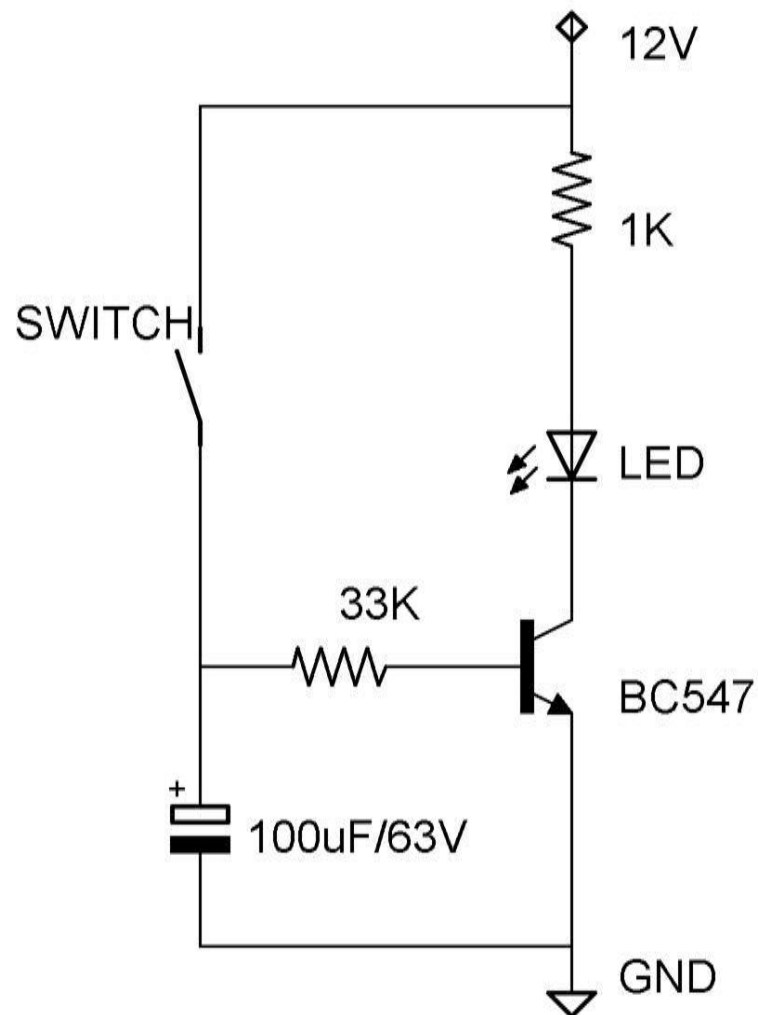
To study Time delay Circuit.

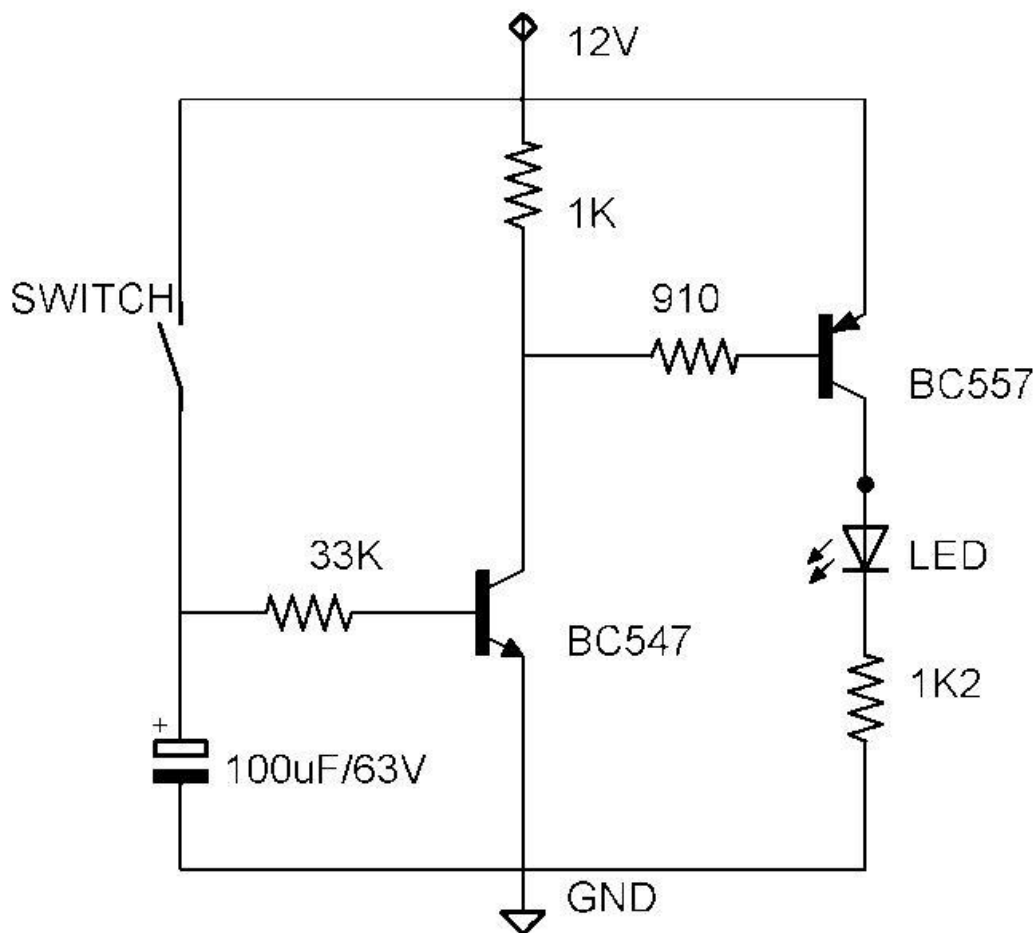
THEORY:-

In many electronic circuit applications a delay of a few seconds or minutes becomes a crucial requirement for ensuring correct operation of the circuit. Without the specified delay the circuit could malfunction or even get damaged. The first circuit diagram shows how transistors and a few other passive components may be connected for acquiring the intended delay timing outputs. The transistor has been provided with the usual base resistor for the current limiting functions. A LED which is used here just indication purposes behaves like the collector load of the circuit. A capacitor, which is the crucial part of the circuit gets the specific position in the circuit, we can see that it's been placed at the other end of the base resistor and not directly to the base of the transistor.

A push button is used to initiate the circuit. On depressing the button momentarily, a positive voltage from the supply line enters the base resistor and switches ON the transistor and subsequently the LED. However in the course of the above action, the capacitor also gets charged fully. On releasing the push button, though the power to the base gets disconnected, the transistor continues to conduct with the aid of the stored energy in the capacitor which now starts discharging its stored charge via the transistor. The LED also stays switched ON until the capacitor gets fully discharged.

The value of the capacitor determines the time delay or for how long the transistor stays in the conducting mode. Along with the capacitor, the value of the base resistor also plays an important role in determining the timing for which the transistor remains switched ON after the push button is released. However the circuit using just one transistor will be able to produce time delays which may range only for a few seconds. By adding one more transistor stage (next figure) the above time delay range can be increased significantly. The addition of another transistor stage increases the sensitivity of the circuit, which enables the use of larger values of the timing resistor thereby enhancing the time delay range of the circuit.

CIRCUIT DIAGRAM:-**Circuit 1**

Circuit 2**PROCEDURE:-**

- Connect the circuit as per circuit diagram.
- Then close the switch, LED will glow immediately.
- Open the switch, LED will remain in ON condition for particular time period & then OFF the LED slowly.

OBSERVATION:-

- Observe the LED when switched ON & OFF position.
- Observe the delay time required to OFF the LED.
- Calculate the delay Time.

**EXPERIMENT
NO. 16**

EXPERIMENT NO: 15

NAME:-

AC Voltage Controller

OBJECTIVE:-

To study AC Voltage Controller by using TRIAC.

THEORY:-

A voltage controller, also called an AC voltage controller or AC regulator is an electronic module based on either thyristors, TRIACs, SCRs or IGBTs, which converts a fixed voltage, fixed frequency alternating current (AC) electrical input supply to obtain variable voltage in output delivered to a resistive load. This varied voltage output is used for dimming street lights, varying heating temperatures in homes or industry, speed control of fans and winding machines and many other applications, in a similar fashion to an autotransformer. Voltage controller modules come under the purview of power electronics. Because they are low-maintenance and very efficient, voltage controllers have largely replaced such modules as magnetic amplifiers and saturable reactors in industrial use.

Modes of Operation

Voltage controllers work in two different ways; either through "on-and-off control" or through "phase control".

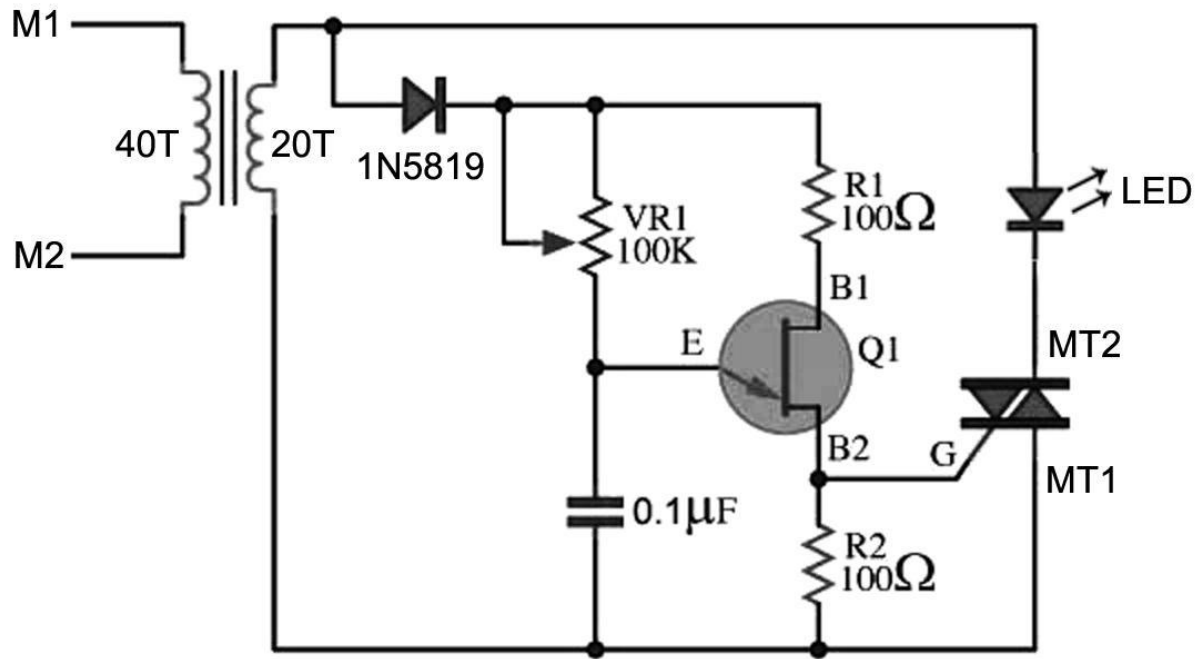
ON-and-OFF control

In an on-and-off controller, thyristors are used to switch on the circuits for a few cycles of voltage and off for certain cycles, thus altering the total RMS voltage value of the output and acting as a high speed AC switch. The rapid switching results in high frequency distortion artifacts which can cause a rise in temperature, and may lead to interference in nearby electronics. Such designs are not practical except in low power applications.

Phase angle control

In phase angle control, thyristors are used to halve the voltage cycle during input. By controlling the phase angle or trigger angle, the output RMS voltage of the load can be varied. The thyristor is turned on for every half-cycle and switched off for each remaining half-cycle. The phase angle is the position at which the thyristor is switched on. TRIACs are often used instead of thyristors to perform the same function for better efficiency. If the load is a combination of resistance and inductance, the current cycle lags the voltage cycle, decreasing overall power output.

CIRCUIT DIAGRAM:-



PROCEDURE:-

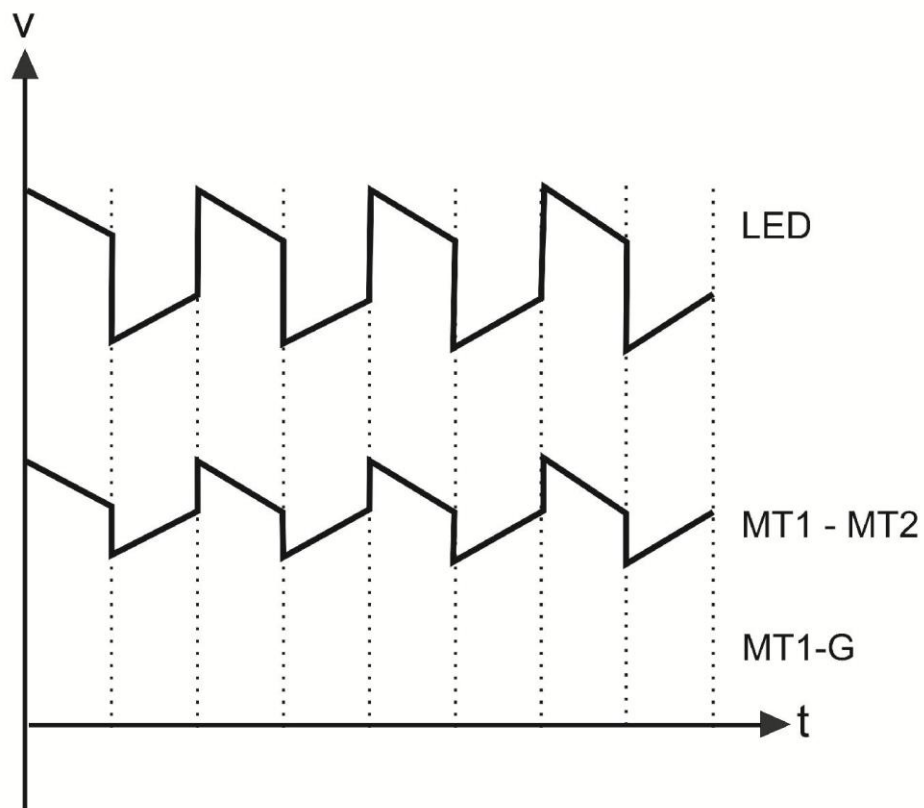
- Connect the circuit as per circuit diagram.
- Keep PWM knob at minimum position so observed sine wave at 20T.
- Initially keep POT Resistance at maximum position. Switch ON the kit.
- LED will not glow, then decrease resistance of POT.
- At particular point LED start glowing.

OBSERVATION:-

- Observe the LED when POT Resistance at maximum & minimum position.
- Observe the waveform across LED.
- Observed the waveforms across different terminals of TRAIC.

WAVEFORMS:-

When POT resistance at maximum position



When POT resistance at minimum position

